Spin–Momentum Impedance and Filtering at a Spin-Coupled Absorbing Boundary

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Abstract

In a separable harmonic waveguide, transverse confinement leaves the longitudinal bulk Hamiltonian unchanged. Earlier simulations nevertheless found that stronger confinement suppresses prompt detection of a spin-1/2 particle at a spin-coupled absorbing boundary. Here we identify the local boundary response underlying this effect. We treat the boundary as an effective detector, identifying the squared-norm loss rate with the detection-time probability density. The incident transverse ground state has a distribution of momenta parallel to the detecting plane, with characteristic scale proportional to $\sqrt{\omega}$, where ω is the confinement frequency. For each parallel momentum, we resolve the boundary relation into two spin channels. At fixed normal wave number and detector parameter, local matching gives these channels equal absorption probabilities but generally different reflection phases. Their common absorption probability decreases with the magnitude of the parallel momentum. Simulations over several observation windows distinguish reduced detection probability from changes in recorded-click timing. The mean detection time capped at horizon T is summarized by the empirical fit $A_T + B_T\sqrt{\omega}$ over the sampled range, with setup- and horizon-dependent coefficients. The local response makes detection sensitive to transverse confinement; propagation, interference, and repeated encounters of the undetected wave shape the detection-time distribution.

1 Introduction

Predicting when a quantum detector clicks requires more than the usual Born rule for position at a prescribed time. Proposed descriptions of arrival and detection times include flux-based rules, detector-free trajectory arrival times, models using complex absorbing potentials or absorbing boundary conditions [1–7], and, more recently, quantum-stroboscopic measurement protocols [8]. An important question is how these descriptions relate to the click-time statistics recorded by a physical detector.

A concrete example is the harmonically confined waveguide studied by Das and Dürr [4]. They calculated spin-dependent first-arrival distributions from Bohmian trajectories without including detector-induced changes in the Pauli evolution. They proposed these intrinsic distributions as experimental benchmarks, expecting a suitable detector to cause only a small disturbance. However, a detector-free arrival time and a detector-present click time need not coincide: the detector can alter the particle's evolution before a click, so its response must be included when predicting the measured distribution [3, 9, 10].

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The relation between these intrinsic arrival times and recorded clicks is constrained by standard quantum measurement theory. For a fixed detector arrangement, click-time probabilities are described by a positive-operator-valued measure (POVM) [3, 5, 10]. In particular, no single fixed detector can reproduce the intrinsic arrival-time distributions of Ref. [4] exactly for all initial spin orientations [9, 10]. These constraints do not, however, determine the response of a particular detector, which must be specified separately [5, 11].

Two effective approaches to specifying such a detector response are absorbing boundary conditions (ABCs) and complex absorbing potentials (CAPs) [5, 6, 11, 12]. A CAP represents absorption within a spatial region through a negative imaginary potential, whereas an ABC represents it by a boundary law at the detecting surface. Both modify the particle's pre-click evolution and can produce reflection as well as absorption [11]. For scalar ABCs, Cavendish and Das found scattering examples in which boundary reflection leads to detection probabilities different from standard scattering-theory predictions even in the far-field limit [13].

We focus on the spin-coupled ABC and identify the squared norm of the subnormalized state with the probability that no detection has occurred. With no other loss channels, its loss rate equals the outward Pauli probability flux integrated over the detecting surface. This nonnegative rate defines the click-time density. The boundary condition and this probability assignment specify an effective detector model without resolving microscopic apparatus degrees of freedom. This detector interpretation is distinct from the numerical uses of absorbers for unbound-particle analysis [14] or radiation conditions [15].

In our recent study [11], we compared these detector models in the same waveguide geometry. In that separable setting, the purely longitudinal CAP and spin-decoupled ABC give detection-time distributions, integrated over detector position, that are independent of transverse confinement. By contrast, the spin-coupled ABC produced confinement-dependent redistribution and delayed structure, while no noticeable dependence on the initial spin orientation was observed in the cases tested. Because transverse confinement leaves the longitudinal bulk Hamiltonian unchanged, the question is how the spin-coupled boundary introduces the observed confinement dependence and how reflected, undetected amplitude contributes to later detection.

Understanding this response helps distinguish detector-induced changes from propagation effects when interpreting click-time measurements and comparing detector models. Beyond this measurement question, momentum-dependent absorption and spin-dependent reflection phases provide a theoretical target for interfaces that select momentum components and modify the spin state of the reflected wave.

The present setup keeps the same waveguide geometry, with harmonic transverse confinement of frequency ω, a reflecting end at $z = 0$, and a detecting plane at $z = L$. The spin-1/2 particle is initially prepared with the transverse ground-state profile and a longitudinal Gaussian wavepacket centered at $z = L/2$, with central momentum along $+z$. At the detecting plane, we adopt the spin-coupled ABC

$$(\boldsymbol{\sigma}\cdot\boldsymbol{\nabla})\Psi\big|_{z=L} = i\kappa(\mathbf{n}\cdot\boldsymbol{\sigma})\,\Psi\big|_{z=L}\,, \qquad \mathbf{n}=\hat{\mathbf{z}},$$

where $\mathbf{n}$ is the outward unit normal, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the vector of Pauli matrices, and $\kappa > 0$ is an inverse-length parameter governing absorption and reflection. For each tangential wave vector $\boldsymbol{\xi}$, parallel to the detecting plane, the Fourier-transformed boundary condition relates the boundary value to its outward normal derivative through a matrix with eigenvalues $i\kappa \pm |\boldsymbol{\xi}|$. At fixed normal wave number and κ, local matching gives equal absorption probabilities and generally different reflection phases in the two channels, while absorption decreases as $|\boldsymbol{\xi}|$ increases.

The incident transverse ground state has a distribution of tangential wave vectors with characteristic magnitude proportional to $\sqrt{\omega}$ in the units used below. Stronger confinement therefore gives greater weight to components that are less efficiently absorbed in the local matching problem at fixed normal wave number and κ. The complete detection-time signal also depends

on transverse-mode coupling, propagation, interference, and repeated encounters of the undetected wave with the detector. We combine local matching with full wavepacket simulations, scalar-boundary comparisons, and lower-boundary controls to determine how tangential boundary coupling affects finite-guide detection statistics. Comparisons at several observation horizons distinguish changes in detection probability from changes in the timing of recorded clicks. Numerical details, supporting derivations, control calculations, and illustrative physical scales are given in the Supplementary Information [16].

2 Model and Detector-Present Observable

We work in units $\hbar = m = 1$. The spinor obeys $i\partial_t\Psi = H\Psi$, with spin-independent bulk differential expression

$$H = -\frac{1}{2}\Delta + \frac{1}{2}\omega^2\left[(x-x_c)^2 + (y-y_c)^2\right], \tag{1}$$

Here (x_c, y_c) is the transverse trap center. The numerical transverse domain has Dirichlet walls at fixed distances in oscillator units; its scaling and resolution are specified in Supplementary Information, Sec. S2. The initial state is factorized as

$$\Psi_0(x, y, z) = \chi_\omega(x, y)\,\phi_0(z)\,\eta, \tag{2}$$

where $\eta \in \mathbb{C}^2$ is a constant normalized spinor and χ_ω is the normalized transverse harmonic-ground-state profile, with oscillator length $\ell_\perp = \omega^{-1/2}$. For the confinement study in a guide of length $L > 0$, ϕ_0 is a longitudinal Gaussian centered at $z_c = L/2$, with central wave number $k_z > 0$ along $+z$ and amplitude-width parameter $\sigma_\parallel$. Its explicit form and width convention are given in Supplementary Information, Sec. S2, Eq. (S5). The guide has a reflecting Dirichlet boundary $\Psi(x, y, 0, t) = 0$ at $z = 0$, and its upper end is the detecting plane

$$\Sigma_L = \{z = L\}. \tag{3}$$

At this plane, we adopt the spin-coupled absorbing boundary condition as a defining assumption of the effective detector model:

$$(\boldsymbol{\sigma}\cdot\boldsymbol{\nabla})\Psi = i\kappa\sigma_z\Psi, \quad \kappa > 0. \tag{4}$$

Equation (4) also arises from the semi-ideal Dirac absorbing-boundary model in the nonrelativistic limit with the boundary-parameter scaling specified in Ref. [17]. This is a limiting relation between idealized models. Derivations of scalar absorbing-boundary detection laws from detector models have also been investigated [18, 19]. Mathematical results for related scalar absorbing-boundary problems are given in Refs. [20, 21]. For sufficiently regular spinor solutions, the Pauli current is

$$\boldsymbol{j}^P = \mathrm{Im}(\Psi^\dagger\boldsymbol{\nabla}\Psi) + \frac{1}{2}\boldsymbol{\nabla}\times(\Psi^\dagger\boldsymbol{\sigma}\Psi). \tag{5}$$

The spinor ABC gives the outward normal probability-current density

$$j_z^P(x, y, L, t) = \kappa\rho(x, y, L, t), \qquad \rho = \Psi^\dagger\Psi. \tag{6}$$

Let $\tau \in [0, \infty]$ denote the random time of the detector click in a single realization, with $\tau = \infty$ if no detection ever occurs. By contrast, $t \geq 0$ is the deterministic evolution time and serves as the integration variable below; $T > 0$ is a fixed observation horizon. For normalized initial data, the model identifies $\|\Psi_t\|^2 = \mathbb{P}(\tau > t)$. With no flux through the nondetecting boundaries, the continuity equation then gives the density of finite click times,

$$g(t;\omega) = \kappa\int_{\Sigma_L}\rho(x, y, L, t)\,dxdy = -\frac{d}{dt}\|\Psi_t\|^2. \tag{7}$$

For a finite observation horizon T, define the detected fraction

$$D_T(\omega) = \int_0^T g(t;\omega)\, dt, \tag{8}$$

and the finite-window restricted mean detection time, with click times capped at T,

$$\mu^*(T;\omega) = \int_0^T S(t;\omega)\, dt, \qquad S(t;\omega) = \|\Psi_t\|^2. \tag{9}$$

Equivalently,

$$\mu^*(T;\omega) = \mathbb{E}[\min(\tau, T)]. \tag{10}$$

For $D_T(\omega) > 0$, let $\overline{\tau}_T^{\mathrm{cond}}(\omega)$ denote the mean click time conditioned on detection by T. Then

$$\mu^*(T;\omega) = D_T(\omega)\, \overline{\tau}_T^{\mathrm{cond}}(\omega) + T[1 - D_T(\omega)]. \tag{11}$$

We use the restricted mean to summarize the finite observation window without extrapolating the unobserved tail. Recorded detections contribute their click times; realizations undetected by T contribute T to this statistic, so $0 \le \mu^*(T;\omega) \le T$. Reporting it alongside D_T and the conditional mean separates changes in detection probability from changes in the timing of recorded clicks. Further details are given in Supplementary Information, Sec. S3.

3 Confinement-dependent detection statistics

Figure 1 shows the detector-present click-time density for representative confinements, with the longitudinal preparation, detector parameter, initial spinor, and guide length fixed. Increasing ω suppresses the prompt peak and reshapes the subsequent oscillatory signal. To compare early and late detections across confinements, we decompose the observation window at $t_{\mathrm{cut}} = 3.6$ for every ω, defining the early window $[0, t_{\mathrm{cut}}]$ and the late window $(t_{\mathrm{cut}}, T]$[1]. Stronger confinement reduces the early detected probability, while the mean click time conditioned on early detection changes only weakly. At $T = 40$, both the fraction of recorded clicks in the late window and their conditional mean time increase across the sampled confinements (Supplementary Information, Secs. S3.2–S3.3 and Table S2).

The lower panel in Fig. 1 resolves the late-time structure; the dashed detector-free Gaussian current in the upper panel marks the corresponding free crossing comparison curve. Numerical settings are given in Supplementary Information, Sec. S2.

[1] This common cutoff is a diagnostic division, not a boundary between first encounters and subsequent returns.

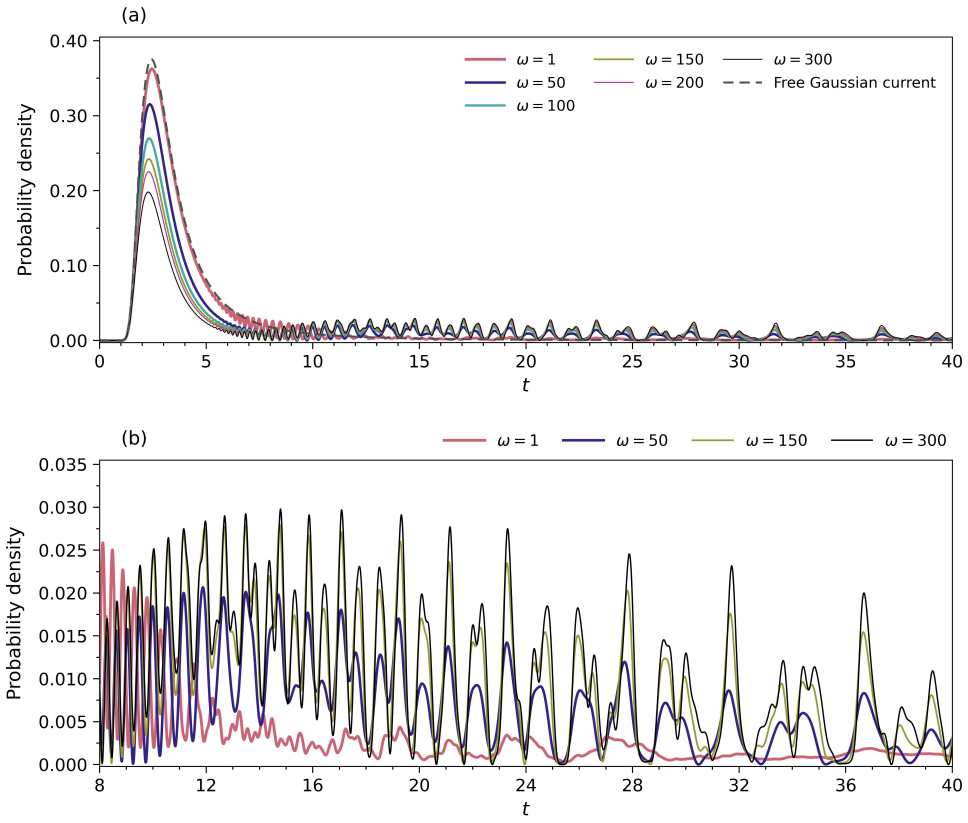
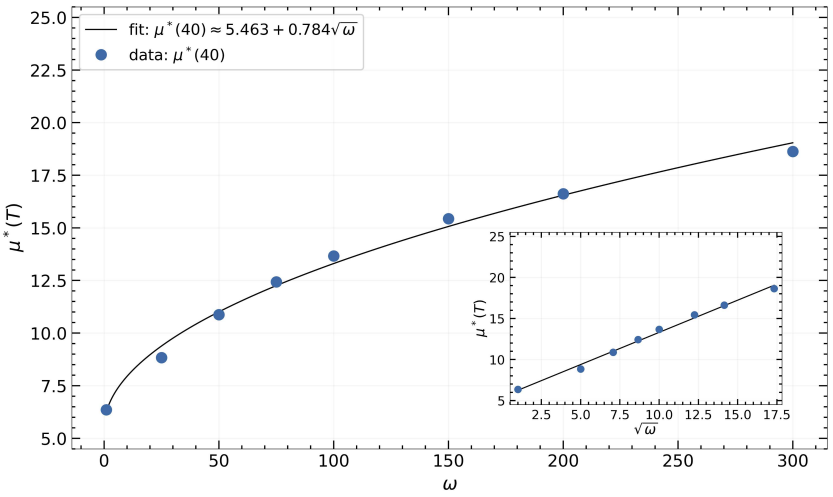


Figure 1: Detector-present click-time density $g(t;\omega)$ for representative confinements, with $L = 20$, $\kappa = k_z = \pi$, $\sigma_{\|} = 0.5$, and $\eta = (1,0)^{\mathsf{T}}$. (a) Profiles for $\omega = 1, 50, 100, 150, 200, 300$ over $0 \le t \le 40$; the black dashed curve is the detector-free Gaussian comparison current. (b) Late-time detail over $8 \le t \le 40$ for $\omega = 1, 50, 150, 300$, with the same colors as in (a). The detector-present curves are not conditioned on detection by $t = 40$; their areas in (a) equal the detected fractions $D_{40}(\omega)$. The choice $\kappa = k_z$ matches the scalar boundary to the central longitudinal plane-wave component. Numerical settings and detected fractions are given in Supplementary Information, Sec. S2 and Table S2, respectively.

Figure 2 summarizes the restricted mean at $T = 40$ for the eight sampled confinements. An unweighted least-squares fit gives

$$\mu^*(40;\omega) \simeq 5.463 + 0.784\sqrt{\omega}. \tag{12}$$

This empirical fit describes the sampled confinements at $T = 40$; extrapolation beyond this range or horizon is not established.



restricted mean detection time, $T = 40$

Figure 2: Restricted mean detection time at $T = 40$ for $\omega \in \{1, 25, 50, 75, 100, 150, 200, 300\}$, with the preparation of Fig. 1. Markers are calculated values; the solid line is the unweighted fit in Eq. (12). The inset shows the same data against $\sqrt{\omega}$. Fit residuals and numerical integration conventions are given in Supplementary Information, Sec. S3.

To distinguish changes in detection probability from changes in the timing of recorded clicks, Fig. 3 compares $D_T(\omega)$, $\bar{\tau}_T^{\mathrm{cond}}(\omega)$, and $\mu^*(T;\omega)$ at $T = 10, 20, 30, 40$, using prefixes of the same recorded evolutions. Across the sampled confinements, D_T decreases and μ^* increases at every displayed horizon. The conditional mean increases with confinement at $T = 20, 30, 40$, but varies weakly and nonmonotonically at $T = 10$. This illustrates Eq. (11): a larger undetected fraction can raise μ^* even when the conditional mean changes little.

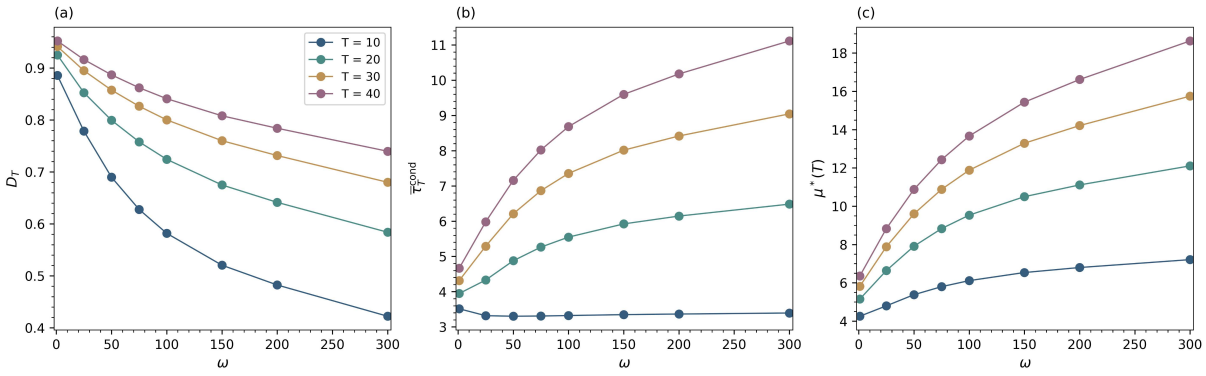


Figure 3: Observation-window dependence for the eight reference confinements: (a) detected fraction $D_T(\omega)$; (b) conditional mean $\bar{\tau}_T^{\mathrm{cond}}(\omega) = M_T(\omega)/D_T(\omega)$, where $M_T(\omega) = \int_0^T t\, g(t;\omega)\, dt$; and (c) restricted mean $\mu^*(T;\omega)$. Each panel shows $T = 10, 20, 30, 40$, evaluated from prefixes of the same recorded evolutions. Markers are calculated values; connecting lines guide the eye.

At $\omega = 300$, about 26% of the probability remains undetected at $T = 40$; the survival contribution $40[1 - D_{40}(300)]$ accounts for about 56% of $\mu^*(40;300)$. An additional long-time calculation at $\omega = 200$ shows that a substantial fraction of the probability undetected at $T = 40$ contributes to later clicks (Supplementary Information, Sec. S9.6 and Fig. S13). The unrestricted mean remains undetermined by these finite-window data.

4 Local Boundary Response and Returns in the Guide

At the detecting plane, part of an incident wave can be absorbed and part reflected. The undetected reflected amplitude can return to the detector after reflection at the lower wall $z = 0$. We separate the analysis into four parts: **(i)** a separable bulk reference for the first encounter; **(ii)** the boundary spin channels and their local reflection and absorption; **(iii)** the auxiliary normal variation of the evolved spin-dependent boundary trace; and **(iv)** lower-boundary controls and a reduced description of subsequent returns to the detecting plane. The local calculation identifies how transverse motion enters the boundary condition, while the complete detection-time distribution also depends on propagation and interference in the finite guide.

(i) A separable first-pass bulk reference. For comparison, consider an auxiliary bulk evolution whose Hamiltonian acts identically on the two spin components, $H_{\mathrm{bulk}} = [H_{xy}(\omega) + H_z]I_2$. Here $H_{xy}(\omega)$ contains the transverse kinetic and harmonic terms, while H_z is the longitudinal kinetic operator. For the factorized initial state in Eq. (2), separability preserves the form

$$\Psi_{\mathrm{fp}}(x, y, z, t) = \chi_\omega(x, y, t)\phi_{\mathrm{fp}}(z, t)\eta, \tag{13}$$

where $\phi_{\mathrm{fp}}(z, 0) = \phi_0(z)$ and the prepared spinor η is unchanged. Here "spin independent" refers to the Hamiltonian, not to the state, which still depends on η. The corresponding Pauli current is

$$\boldsymbol{j}^P_{\mathrm{fp}} = \boldsymbol{j}^{\mathrm{conv}}_{\mathrm{fp}} + \frac{1}{2}\boldsymbol{\nabla} \times (\rho_{\mathrm{fp}}\boldsymbol{s}_\eta), \quad \boldsymbol{s}_\eta := \eta^\dagger \boldsymbol{\sigma}\eta, \quad \rho_{\mathrm{fp}} := |\Psi_{\mathrm{fp}}|^2. \tag{14}$$

For this auxiliary separable field, $\mathbf{s}_\eta$ is spatially constant, so the normal spin-curl term is a transverse divergence. By Stokes' theorem on the transverse cross-section, together with the lateral Dirichlet conditions, its cross-section integral vanishes. Hence, using transverse normalization, the flux through an interior plane $\Sigma_b = \{z = z_b\}$, with $z_b < L$, satisfies

$$\int_{\Sigma_b} j^P_{\mathrm{fp},z}\, dA = \int_{\Sigma_b} \mathrm{Im}(\Psi^\dagger_{\mathrm{fp}}\partial_z\Psi_{\mathrm{fp}})\, dA = \mathrm{Im}(\phi^*_{\mathrm{fp}}\partial_z\phi_{\mathrm{fp}})(z_b, t). \tag{15}$$

This plane-integrated flux is independent of ω and of the prepared spin orientation, although the local current may depend on spin. Note that, the full spinor-ABC evolution need not preserve the factorized form. Details of the auxiliary bulk calculation are given in Supplementary Information, Sec. S4.

(ii) Boundary spin channels, reflection, and absorption. The separable reference isolates the longitudinal bulk flux. In the detector-present problem, the boundary condition couples the normal derivative to transverse motion and spin. At $z = L$, the spinor ABC in Eq. (4) reads

$$\begin{aligned} \partial_z\psi_\uparrow &= i\kappa\,\psi_\uparrow - (\partial_x - i\partial_y)\psi_\downarrow, \\ \partial_z\psi_\downarrow &= i\kappa\,\psi_\downarrow + (\partial_x + i\partial_y)\psi_\uparrow. \end{aligned} \tag{16}$$

Equivalently,

$$\partial_z\Psi|_{z=L} = C\,\Psi|_{z=L}, \quad C = \begin{pmatrix} i\kappa & -D_- \\ D_+ & i\kappa \end{pmatrix}, \quad D_\pm = \partial_x \pm i\partial_y. \tag{17}$$

Let $\widehat{\Psi}(\boldsymbol{\xi}, z, t)$ be the Fourier transform of the spinor in the tangential coordinates (x, y), where $\boldsymbol{\xi} = (\xi_x, \xi_y)$ is the corresponding tangential wave vector. At the detecting plane,

$$\partial_z\widehat{\Psi}(\boldsymbol{\xi}, L, t) = \mathcal{C}(\boldsymbol{\xi})\widehat{\Psi}(\boldsymbol{\xi}, L, t). \tag{18}$$

Therefore, the matrix $\mathcal{C}(\boldsymbol{\xi})$, the tangential boundary symbol, maps the boundary value of each tangential Fourier component to its outward normal derivative. From Eq. (17),

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + (\hat{z} \times \boldsymbol{\xi}) \cdot \boldsymbol{\sigma}. \tag{19}$$

For $R := |\boldsymbol{\xi}| > 0$, write

$$\Gamma(\boldsymbol{\xi}) = \frac{(\hat{z} \times \boldsymbol{\xi}) \cdot \sigma}{R}, \qquad \Gamma(\boldsymbol{\xi})^\dagger = \Gamma(\boldsymbol{\xi}), \qquad \Gamma(\boldsymbol{\xi})^2 = I. \tag{20}$$

Therefore Eq. (19) can be rewritten as

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + R\Gamma(\boldsymbol{\xi}). \tag{21}$$

The two tangential spin–momentum projectors are then

$$\Pi_\pm(\boldsymbol{\xi}) = \frac{1}{2}\big(I \pm \Gamma(\boldsymbol{\xi})\big). \tag{22}$$

The projectors $\Pi_\pm(\boldsymbol{\xi})$ select the two particle-spin orientations along the tangential axis $\mathbf{n}_{\boldsymbol{\xi}} = (\hat{\mathbf{z}} \times \boldsymbol{\xi})/R$, with eigenvalues ± 1 of $\mathbf{n}_{\boldsymbol{\xi}} \cdot \boldsymbol{\sigma}$. At $R = 0$, $\mathcal{C}(0) = i\kappa I$ and the branches are degenerate. A confined state contributes a distribution of tangential wave vectors. The projectors act componentwise on that distribution; the boundary trace is not replaced by a single momentum eigenstate. The harmonic bulk evolution generally couples these Fourier components. The eigenvalues of $\mathcal{C}(\boldsymbol{\xi})$ are

$$\lambda_\pm(\boldsymbol{\xi}) = i\kappa \pm R. \tag{23}$$

Local reflection and absorption. The scattering meaning of these eigenvalues follows from a local plane-wave matching problem at the detecting plane. Fix the tangential wave vector $\boldsymbol{\xi}$ and a positive normal wave number k. For either boundary-spin branch, write the normal amplitude as

$$\psi_\pm(z) = e^{ik(z-L)} + r_\pm e^{-ik(z-L)}, \qquad z \le L. \tag{24}$$

Imposing the corresponding boundary relation gives

$$ik(1 - r_\pm) = (i\kappa \pm R)(1 + r_\pm), \tag{25}$$

and hence

$$r_\pm(k, R) = \frac{i(k-\kappa) \mp R}{i(k+\kappa) \pm R}, \qquad \mathcal{A}_\pm(k, R) = 1 - |r_\pm(k, R)|^2 = \frac{4k\kappa}{(k+\kappa)^2 + R^2}. \tag{26}$$

Here $\mathcal{A}_\pm$ is the fraction of incident normal flux absorbed at the boundary. For the unit incident amplitude used above, the incident normal flux is k, so the detector law $j_z^P = \kappa\rho$ gives $\mathcal{A}_\pm = (\kappa/k)|1 + r_\pm|^2$. Therefore the tangential-momentum dependence enters through the boundary density determined by matching. At fixed k and κ, increasing R reduces absorption and increases reflection equally in the two branches. Their different reflection phases modify the relative phase of a coherent superposition of the boundary-spin channels.

We use "boundary impedance" for the matrix relation between the boundary value and its outward normal derivative. Here "spin–momentum filtering" combines momentum-dependent absorption with spin-dependent phase changes upon reflection. Replacing the spin-coupled boundary law by the componentwise scalar condition $\partial_z \Psi = i\kappa\Psi$ gives $r_{\mathrm{sc}}(k) = (k-\kappa)/(k+\kappa)$, which is independent of tangential momentum at fixed k and vanishes only at $k = \kappa$. The scalar ABC controls show nearly confinement-independent finite-window statistics in the tested cases, in contrast with the spinor response (Supplementary Information, Sec. S3.3).

These coefficients in Eq. (26) describe the frozen local matching problem. The complete confined-wavepacket signal also involves transverse-mode coupling, propagation, and interference. We next examine the same boundary relation through the local normal variation of the evolved boundary trace.

(iii) Auxiliary normal response and moment variation. To complement the local scattering calculation, we use the evolved boundary trace $W(\boldsymbol{\xi}, t) = \widehat{\Psi}(\boldsymbol{\xi}, L, t)$ to probe the first

inward variation of a normalized time moment. This auxiliary construction diagnoses the local normal response of the evolved state rather than providing an independent prediction of the detection-time distribution. At each fixed time and for small $\epsilon = L - z > 0$, the field can be written as the homogeneous normal continuation plus a remainder:

$$\widehat{\Psi}(\boldsymbol{\xi}, L-\epsilon, t) = e^{-i\kappa\epsilon} B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W(\boldsymbol{\xi}, t) + \widehat{\mathcal{R}}_{\mathrm{Duh}}, \tag{27}$$

where

$$B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) := e^{-\epsilon[C(\boldsymbol{\xi}) - i\kappa I]} = e^{-R\epsilon}\Pi_+(\boldsymbol{\xi}) + e^{R\epsilon}\Pi_-(\boldsymbol{\xi}). \tag{28}$$

Integrating the auxiliary homogeneous normal relation inward gives $e^{-\epsilon\mathcal{C}(\boldsymbol{\xi})}W(\boldsymbol{\xi}, t)$. Equation (28) resolves the auxiliary inward continuation into an exponentially decaying factor $e^{-R\epsilon}\Pi_+ W$ and an exponentially growing factor $e^{R\epsilon}\Pi_- W$. The former is the evanescent factor of this auxiliary continuation; the latter does not represent physical gain[2]. The remainder $\widehat{R}_{\mathrm{Duh}}$ accounts for the difference from the true time-dependent Schrödinger solution. Under the regularity assumptions stated in the Supplementary Information, Sec. S6.1, it is $O(\epsilon^2)$ modewise. After removal of the scalar phase, the boundary values and first normal variation are

$$B_{\mathrm{br}}(0, \boldsymbol{\xi}) = I, \qquad \partial_\epsilon B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi})|_{\epsilon=0} = -R\,\Gamma(\boldsymbol{\xi}). \tag{29}$$

Equation (29) gives the first inward normal variation of each phase-removed tangential Fourier component at the detecting plane. For the incident transverse ground state, $\ell_\perp = \omega^{-1/2}$, giving the characteristic preparation scale

$$R = |\boldsymbol{\xi}| \sim \ell_\perp^{-1} \sim \sqrt{\omega}. \tag{30}$$

The evolved boundary trace $W(\boldsymbol{\xi}, t)$ is determined by the detector-present dynamics and need not retain this incident momentum distribution. To quantify its local normal response, restrict the time integrals below to $0 \le t \le T$, assume $D_T > 0$, and define the auxiliary measure

$$d\mu^{\mathrm{bl}}_{\omega,\epsilon}(\boldsymbol{\xi}, t) := |B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W(\boldsymbol{\xi}, t)|^2 \, d^2\xi \, dt. \tag{31}$$

This is not a click-time distribution at $z = L - \epsilon$: the variable t is only the time label of the boundary trace $W(\boldsymbol{\xi}, t)$, and the physical detector observable remains the detector flux at $z = L$. Here bl stands for "boundary layer" and labels the auxiliary construction based on the homogeneous inward continuation. Since the scalar phase drops out of densities and the Duhamel term has no first-order contribution at the detecting plane, the first variation is computed from the homogeneous branch probe. With $B_{\mathrm{br}} = e^{-\epsilon R\Gamma}$,

$$|B_{\mathrm{br}} W|^2 = W^\dagger e^{-2\epsilon R\Gamma} W = |W|^2 b_{\omega,\epsilon}, \tag{32}$$

where

$$b_{\omega,0} = 1, \qquad \partial_\epsilon b_{\omega,\epsilon}|_{\epsilon=0} = R a_\omega, \tag{33}$$

and

$$a_\omega(\boldsymbol{\xi}, t) := -2\frac{W^\dagger \Gamma(\boldsymbol{\xi}) W}{W^\dagger W}, \qquad |a_\omega| \le 2. \tag{34}$$

The factor a_ω is minus twice the normalized + versus − branch imbalance of the boundary trace. The ratio is evaluated where $W^\dagger W > 0$. We set $a_\omega = 0$ where $W = 0$ or $R = 0$; these conventions do not change the weighted expressions. Let $F(\boldsymbol{\xi}, t)$ be a real-valued test function, independent of the auxiliary depth ϵ, for which the averages below are finite. Define the normalized measures

[2]The factors $e^{\mp R\epsilon}$ describe the branch-dependent normal amplitude response to tangential coupling, while the same impedance terms $\pm R$ determine the reflection phases through local matching. Thus the auxiliary evanescent normal response and scattering phases share a common boundary impedance, without establishing a finite region of exponential decay in the full wavepacket.

$d\nu^{\mathrm{bl}}_{\omega,\epsilon} = d\mu^{\mathrm{bl}}_{\omega,\epsilon} / \int d\mu^{\mathrm{bl}}_{\omega,\epsilon}$ and $d\nu^{(0)}_{\omega} = d\nu^{\mathrm{bl}}_{\omega,0}$. At $\epsilon = 0$, Parseval's identity and the boundary flux law imply that the temporal marginal of $\nu^{(0)}_{\omega}$ is $g(t;\omega)/D_T(\omega)$. Consequently, its time moment equals the conditional mean introduced in Sec. 2,

$$\mathbb{E}_{\nu^{(0)}_{\omega}}[t] = \overline{\tau}^{\mathrm{cond}}_{T}(\omega). \tag{35}$$

The average of F is then

$$\mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[F] = \frac{\mathbb{E}_{\nu^{(0)}_{\omega}}[F b_{\omega,\epsilon}]}{\mathbb{E}_{\nu^{(0)}_{\omega}}[b_{\omega,\epsilon}]}. \tag{36}$$

Assuming that the auxiliary measures are normalizable near $\epsilon = 0^+$ and that differentiation may be interchanged with the integrals, differentiation at $\epsilon = 0$ gives

$$\partial_\epsilon \mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[F]\big|_{\epsilon=0} = \mathrm{Cov}_{\nu^{(0)}_{\omega}}(F, R a_\omega). \tag{37}$$

Taking $F(\boldsymbol{\xi}, t) = t$ gives the first inward variation of the normalized time moment, $\Lambda_\omega := \partial_\epsilon \mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[t]\Big|_{\epsilon=0}$:

$$\Lambda_\omega = \mathrm{Cov}_{\nu^{(0)}_{\omega}}(t, R a_\omega). \tag{38}$$

Define the dimensionless tangential radius $s = R/\sqrt{\omega}$. Then

$$\Lambda_\omega = \sqrt{\omega}\ \mathrm{Cov}_{\nu^{(0)}_{\omega}}(t, s a_\omega) =: \sqrt{\omega}\, \beta^{\mathrm{bl}}_{\omega}. \tag{39}$$

Equation (39) separates the oscillator wave-number scale from the trace-dependent covariance $\beta^{\mathrm{bl}}_{\omega}$. For a fixed observation window, $|a_\omega| \le 2$ gives

$$|\beta^{\mathrm{bl}}_{\omega}| \le 2T \sqrt{\mathbb{E}_{\nu^{(0)}_{\omega}}[s^2]} \tag{40}$$

This bound constrains the magnitude of the auxiliary response coefficient, but determines neither its sign nor the restricted-mean fit.

Complementary diagnostics resolve the tangential boundary branches and the particle-spin content associated with detection and survival. The integrated detector z-polarization is the spin-up minus spin-down contribution to detector loss over $[0, T]$, divided by the total detector loss; the survivor polarization is the normalized expectation of σ_z in the state remaining inside the guide at T. At $\omega = 300$ and $T = 40$, the tested spatial and time-step refinements at fixed numerical boundary positions preserve the integrated +-branch predominance and the detector–survivor z-polarization contrast. Detailed rate profiles and auxiliary boundary-weighted diagnostics, especially the covariance, remain resolution-sensitive (Supplementary Information, Secs. S6.2–S6.3 and Table S5).

(iv) Lower-boundary controls and subsequent returns. The reflected, undetected wave carries the momentum and spin dependence of the boundary response into its subsequent propagation and interference. After reflection at the lower wall, it can reach the detector again. We call the influence of earlier boundary encounters on the later detector signal *finite-guide memory*.

To test the role of the lower wall, we displace it at $\omega = 200$ while keeping the preparation, source and detector positions, and spatial resolution fixed. Moving the wall farther away reduces D_{40} and increases the late-window conditional mean, while the early detected probability agrees at the reported precision. At the same confinement, replacing the lower Dirichlet wall by a partially absorbing scalar ABC, with the upper detector unchanged, reduces the upper-detection probability in $(3.6, 40]$ from 0.438 to 0.130, with lower escape recorded separately. The window-integrated +-branch fraction and detector-associated z-polarization remain nearly unchanged,

while the final in-guide z-polarization remains close to unity. These controls establish lower-boundary sensitivity of the late signal without assigning individual oscillations to particular round trips (Supplementary Information, Secs. S9.4–S9.5, Tables S11–S12, and Figs. S11–S12).

Supplementary Information, Sec. S7, separates exact near-detector probability balance from an assumed scalar return-kernel closure. Under its stated assumptions, the closure relates the normalized first moments of the untruncated fluxes through an effective return weight and mean lag. It is not calibrated to the simulations and does not determine the reported finite-window statistics. The reported detection-time distributions are obtained from the full spinor-ABC evolution.

5 Conclusion

We have identified the local boundary response underlying the confinement dependence previously reported for the spin-coupled absorbing boundary condition [11]. Local plane-wave matching in the two spin eigenchannels of the boundary symbol gives equal absorption probabilities but generally different reflection phases at the same incident wave vector and detector parameter. At fixed normal wave number and detector parameter, absorption decreases with the magnitude of the tangential momentum, whose characteristic scale in the incident transverse ground state grows as $\sqrt{\omega}$. This explains how confinement enters the local detector response, while the complete click-time distribution also depends on propagation and interference in the guide.

In the reference simulations, stronger confinement reduces the probability of early detection while changing the mean time conditioned on early detection only weakly. At the longer observation horizons $T = 20, 30, 40$, the conditional mean of recorded clicks also increases with confinement. The increase of the restricted mean therefore combines changes in recorded-click timing with a larger undetected fraction at the horizon. The empirical square-root fit describes the sampled confinement range, with setup- and horizon-dependent coefficients; the unrestricted mean remains undetermined by the finite-window data.

Comparisons with a scalar ABC at the detecting plane support the role of tangential spin coupling. Fixed-source lower-boundary controls show that late detection depends on the guide termination. At $\omega = 200$, replacing the reflecting lower wall with a partially absorbing scalar ABC substantially reduces late detections at the unchanged upper spin-coupled detector, while the principal branch and spin-polarization contrasts persist.

The restricted-mean trend persists across the tested detector parameters, longitudinal wave numbers, packet widths, and guide lengths. Selected bulk-spin couplings and an excited chiral preparation also preserve the confinement contrast at the two tested frequencies. The principal branch and spin contrasts survive the tested numerical refinements, although detailed time profiles and auxiliary boundary diagnostics remain resolution-sensitive (Supplementary Information, Secs. S6.3, S8.1 and S9).

The complex entrance response provides a calibration target for absorbing layers and a possible basis for momentum-selective absorption and spin-dependent phase control of reflected matter waves (Supplementary Information, Sec. S8.2). Optical analogues could be explored using polarization-dependent wavefront control and wave-vector-dependent polarization selection [23, 24]. Such applications require interfaces that reproduce the relevant amplitudes and relative phases over a finite momentum range. Matching an auxiliary stationary response alone does not establish agreement of the full detection-time distributions.

Further work is needed to develop a microscopic detector model and test a physical implementation. A separate extension is to investigate how an unmonitored transverse environment changes the populations and coherences reaching the detector, while detector clicks remain governed by the absorbing boundary.

Declarations and acknowledgments

Data availability. Selected processed numerical data and representative density animations are publicly available at `https://github.com/jloOop/Boundary-Spin-Momentum-Filtering-Data`. Additional numerical data supporting the results reported in this study are available from the corresponding author on reasonable request.

Code availability. Selected Python scripts for the simulations, diagnostics, and post-processing are available in the same repository. The code and parameters required to reproduce the reported calculations are available from the corresponding author on reasonable request. The solver follows the numerical framework described in Appendix A of Ref. [11], with the boundary implementation specified in Supplementary Information, Sec. S2.

Competing interests. The author declares no competing interests.

Author contributions. A.J. is the sole author of this work and conceived the study, developed the theoretical analysis, implemented and validated the simulations, analyzed the data, prepared the figures and repository, and wrote the manuscript and Supplementary Information.

Declaration of AI-assisted tools. ChatGPT (OpenAI) was used to assist with drafting and revising the manuscript and Supplementary Information, mathematical and numerical consistency checks, and the development and debugging of Python code. The author reviewed the AI-assisted material, checked the code, calculations, figures, and references, and takes full responsibility for the methods, results, and conclusions.

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Supplementary Information for "Spin–Momentum Impedance and Filtering at a Spin-Coupled Absorbing Boundary"

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S1. SCOPE, NOTATION, AND RELATION TO THE MAIN TEXT

This Supplement specifies the numerical model and develops the boundary analysis of the confinement-dependent detection statistics reported in Ref. [11] of the main text. We adopt the spinor absorbing boundary condition (ABC) as an effective detector model: the outward detector flux, equal to the loss rate of $\|\Psi_t\|^2$, is the detection-time probability density. Spin–momentum filtering denotes momentum-dependent absorption and spin-dependent reflection phases; the two local spin channels have equal absorption probabilities at fixed incident wave vector.

Section S2 gives the numerical specification; S3 separates finite-window detection probability from detection timing. Sections S4–S5 derive the separable bulk reference and local boundary response, and S6 reports boundary and spin diagnostics and refinement checks. Section S7 gives probability balance and a reduced model of repeated encounters. Sections S8–S10 collect controls, test robustness via parameter scans, and give physical scales.

S2. NUMERICAL MODEL AND GRID SCALING

In units $\hbar = m = 1$, we evolve the two-component spinor $\Psi = (\psi_\uparrow, \psi_\downarrow)^T$ in the domain

$$\Omega_\omega = [0, L_x(\omega)] \times [0, L_y(\omega)] \times [0, L], \tag{S1}$$

according to

$$i\partial_t \Psi = \left[-\frac{1}{2}\Delta + \frac{1}{2}\omega^2 \big((x - x_c)^2 + (y - y_c)^2 \big) \right] \Psi, \qquad \Psi = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}. \tag{S2}$$

The initial state is a normalized factorized packet,

$$\Psi_0(x, y, z) = \chi_\omega(x, y)\, \phi_0(z)\, \eta, \tag{S3}$$

where the transverse factor is the harmonic ground-state profile, restricted and normalized in the finite transverse box,

$$\chi_\omega(x, y) = \mathcal{N}_\perp(\omega) \exp\left[-\frac{\omega}{2} \big((x - x_c)^2 + (y - y_c)^2 \big) \right], \tag{S4}$$

and the longitudinal factor is the right-moving Gaussian

$$\phi_0(z) = \mathcal{N}_z \exp\left[-\frac{(z - z_c)^2}{2\sigma_\parallel^2} + ik_z(z - z_c) \right], \qquad 0 < z < L. \tag{S5}$$

The constants $\mathcal{N}_\perp(\omega)$ and $\mathcal{N}_z$ normalize the factors on their finite numerical domains; equivalently, the full discrete state is normalized to unit norm before time evolution. The longitudinal Gaussian has

positive central wave number k_z and amplitude-width parameter $\sigma_\parallel$; its untruncated standard deviations are $\Delta z = \sigma_\parallel/\sqrt{2}$ and $\Delta k_z = 1/(\sqrt{2}\,\sigma_\parallel)$. The constant spinor is

$$\eta = \big(\cos(\theta/2), e^{i\phi}\sin(\theta/2)\big)^T, \tag{S6}$$

with $\theta = \phi = 0$ for the reference sweep. Across that sweep, the longitudinal preparation, detector parameter, initial spinor, and spatial grid prescription are fixed as listed in Table S1. The discrete initialization is specified below.

Boundary conditions and observables. The lower face $z = 0$ is a reflecting Dirichlet wall; the artificial transverse walls are also Dirichlet. At the detecting plane $\Sigma_L = \{z = L\}$, the spinor ABC and its outward Pauli current are

$$(\boldsymbol{\sigma}\cdot\boldsymbol{\nabla})\Psi = i\kappa\sigma_z\Psi, \qquad j_z^{\mathrm{P}}(x,y,L,t) = \kappa\Psi^\dagger\Psi \geq 0, \tag{S7}$$

where $\boldsymbol{\sigma}$ denotes the Pauli matrices and $\kappa > 0$. For normalized initial data, the survival probability and detection-time density are

$$S(t;\omega) = \|\Psi_t\|^2, \qquad g(t;\omega) = \kappa\int_{\Sigma_L}\Psi^\dagger\Psi\,dx\,dy = -\partial_t S(t;\omega). \tag{S8}$$

For an observation horizon T_{obs}, we use

$$\begin{aligned} D_{T_{\mathrm{obs}}}(\omega) &= \int_0^{T_{\mathrm{obs}}} g(t;\omega)\,dt, \\ \mu^*(T_{\mathrm{obs}}) &= \mathbb{E}[\min(\tau, T_{\mathrm{obs}})] = \int_0^{T_{\mathrm{obs}}} S(t)\,dt. \end{aligned} \tag{S9}$$

where τ is the detector click time. The restricted mean μ^* includes realizations undetected at the horizon without extrapolating the unobserved tail.

Discretization and recorded statistics. The Cartesian evolution uses the Crank–Nicolson and GMRES framework of Ref. [11], with a centered ghost-point approximation replacing the earlier one-sided normal-boundary closure. The continuum spinor ABC and surface detector-loss law are unchanged. The centered derivative is second-order accurate for smooth fields, and the compatible discrete norm assigns half weight to the terminal longitudinal plane. Sensitivity of the computed observables is assessed separately in Sec. S6.3. The compatible discrete norm is given in Eq. (S12).

Table S1 gives the preparation and numerical settings. The transverse box scales with the oscillator length $\ell_\perp = \omega^{-1/2}$. With $h_\perp = h_x = h_y$,

$$L_x = L_y = \ell_{\mathrm{box}}\ell_\perp, \qquad \frac{h_\perp}{\ell_\perp} = \frac{\ell_{\mathrm{box}}}{N_x}, \qquad \ell_{\mathrm{box}} = 10, \quad N_x = N_y. \tag{S10}$$

Thus the nominal wall distances and transverse resolution in oscillator units are fixed across the sweep.

TABLE S1. Physical preparation and numerical settings for the eight reference evolutions used in the main text and Sec. S6. Units are $\hbar = m = 1$.

Quantity	Value or prescription
Confinement frequencies	$\omega \in \{1, 25, 50, 75, 100, 150, 200, 300\}$
Time-step	$\Delta t \in \{10, 8, 7, 5, 3, 2.5, 2, 1.5\} \times 10^{-4}$,
Nominal box sizes	$L_x = L_y = 10/\sqrt{\omega}$; $L = 20$
Trap and source centers	$x_c = L_x/2$, $y_c = L_y/2$; $z_c = 10$
Longitudinal preparation	$k_z = \pi$, $\sigma_\parallel = 0.5$
Detector and initial spin	$\kappa = \pi$; $\eta = (1, 0)^\mathsf{T}$ $(\theta = \varphi = 0)$
Stored spatial grid	$N_x = N_y = 100$, $N_z = 1500$; nominal upper endpoints excluded
Mesh spacings	$h_x = h_y \simeq 0.1/\sqrt{\omega}$; $h_z \simeq 0.013$
Numerical detecting plane	$z_D = z_{N_z-1} \simeq 19.987$
Evolution scheme	Cartesian finite differences; Crank–Nicolson (Cayley) steps
Linear solver	Jacobi-preconditioned GMRES; rtol = 10^{-8}, atol = 0, restart 30, `maxiter`= 1000
Numerical precision	Coordinates: `float32`; stored state and matrix assembly: `complex64`; linear solves and additional spin/branch diagnostics: `complex128`/`float64`

The coordinate arrays exclude the nominal upper endpoints, and the numerical detecting plane is the last stored longitudinal plane, $z_D = z_{N_z-1}$. Unless otherwise stated, the spinor ABC is implemented through the centered ghost-point relation

$$\frac{\Psi_{N+1} - \Psi_{N-1}}{2h_z} = \mathsf{C}_h \Psi_N, \qquad N = N_z - 1, \tag{S11}$$

where Ψ_N is the grid spinor on the numerical detecting plane and C_h is the centered tangential finite-difference form of the boundary operator in main-text Eq. (17). This relation eliminates the ghost value Ψ_{N+1} from the longitudinal Laplacian and gives a second-order approximation to the normal derivative for smooth fields.

Before evolution, the sampled Gaussian is set to zero on the stored lateral boundary planes and at $z = 0$, then normalized. These Dirichlet values are enforced after every step, without renormalizing the evolved state. The unmodified product Gaussian is not assumed to satisfy the continuum spinor ABC at $t = 0$. The numerical squared norm is

$$P(t) = h_x h_y h_z \sum_{i,j} \left[\sum_{k=0}^{N_z-2} \Psi_{ijk}^\dagger \Psi_{ijk} + \frac{1}{2} \Psi_{ij,N_z-1}^\dagger \Psi_{ij,N_z-1} \right]. \tag{S12}$$

The terminal half weight makes this discrete norm compatible with the centered boundary closure. For the reference geometry, the unrounded stencil and exact Crank–Nicolson solves give a decrease

of P equal to the detector flux integrated over the detecting plane, evaluated on the midpoint spinor $(\Psi^{n+1}+\Psi^n)/2$, multiplied by the step duration. The same volume weights are used for normalization, survival probabilities, and survivor-spin matrices.

Mass sums are accumulated in double precision, and probability statistics are normalized by the recorded launch norm $P_0 = P(0)$. Detector-event probabilities are recorded separately from casting and boundary-clamp corrections, which are excluded from detector events. Each reference evolution reaches $T_{\max} = 40$, with $T = 10, 20, 30, 40$ evaluated from the same history. Event probabilities and stored-state masses are recorded at every accepted step, plane currents every 20 steps, and spin/branch diagnostics at nominal intervals of 0.02. All integrations use the actual recorded times.

Refinement checks for the detection statistics, detector rates, and boundary-branch diagnostics at $\omega = 300$ and $T = 40$ are reported in Sec. S6.3.

S3. FINITE-WINDOW DETECTION STATISTICS

This section separates the probability of detection by a finite horizon from the timing of the recorded clicks. We use the eight reference evolutions of Table S1 and report the $T = 40$ statistics in Table S2.

S3.1. Conditional detection-time statistics and restricted-mean identity

Using g, S, D_T, and μ^* defined in Sec. S2, the density conditioned on detection by T, for $D_T(\omega) > 0$, and its mean are

$$\begin{aligned} f_T(t;\omega) &= \frac{g(t;\omega)}{D_T(\omega)}, \qquad 0 \le t \le T, \\ \bar{\tau}_T^{\mathrm{cond}}(\omega) &= \int_0^T t\, f_T(t;\omega)\, dt = \frac{M_T(\omega)}{D_T(\omega)}, \\ M_T(\omega) &= \int_0^T t\, g(t;\omega)\, dt. \end{aligned} \tag{S13}$$

Recorded detections contribute their click times, while realizations undetected at T contribute the horizon itself,

$$\begin{aligned} \mu^*(T;\omega) &= \mathbb{P}(\tau \le T)\, \mathbb{E}[\tau \mid \tau \le T] \\ &\quad + \mathbb{P}(\tau > T)\, \mathbb{E}[T \mid \tau > T] \\ &= \mathbb{P}(\tau \le T)\, \mathbb{E}[\tau \mid \tau \le T] + T\, \mathbb{P}(\tau > T) \\ &= D_T(\omega)\, \bar{\tau}_T^{\mathrm{cond}}(\omega) + [1 - D_T(\omega)]\, T. \end{aligned} \tag{S14}$$

Equivalently,

$$\bar{\tau}_T^{\mathrm{cond}}(\omega) = \frac{\mu^*(T;\omega) - [1 - D_T(\omega)]T}{D_T(\omega)}. \tag{S15}$$

Since $0 \le \mu^*(T;\omega) \le T$, an affine fit in $\sqrt{\omega}$ with nonzero slope can describe only a finite confinement range at a specified horizon.

For the spinor reference evolutions, D_T and M_T are evaluated from recorded Crank–Nicolson boundary probabilities, using a constant rate within each accepted step and interval overlap at window endpoints. The restricted mean is independently integrated from the piecewise-linear stored survival history. All probability quantities are divided by the recorded launch norm. The largest absolute detector–survival budget discrepancy over the eight histories is below 3.68×10^{-7}. The largest absolute residual in Eq. (S14), evaluated at $T = 10, 20, 30, 40$, is below 1.18×10^{-5}. These measure consistency of the recorded statistics, not discretization accuracy.

S3.2. Common diagnostic cutoff

The early/late split uses one cutoff time t_{cut}, common to all values of ω. The cutoff is not fitted from the detector-flux data and is not chosen separately for each run. It is used only to define a fixed diagnostic window for comparing the early and late contributions to the detector signal. Let

$$\Sigma_i = \{z = z_i\} \tag{S16}$$

be an interior monitoring plane below the detector. From the main upward-flux peak times on two such planes, we estimate an effective axial speed $v_{\mathrm{est}}(\omega)$. If $t^{(i)}_{\mathrm{pk,up}}(\omega)$ is the main upward peak time on Σ_i, then the estimated arrival time at Σ_i of the signal reflected from the detecting plane is

$$t^{(i)}_{\mathrm{return}}(\omega) = t^{(i)}_{\mathrm{pk,up}}(\omega) + \frac{2d^{(i)}_R}{v_{\mathrm{est}}(\omega)}, \qquad d^{(i)}_R = z_{\mathrm{detector}} - z_i. \tag{S17}$$

A subsequent signal reflected from the lower wall would require an additional round trip from the monitoring plane to the bottom and back. We estimate the corresponding cycle-completion time by

$$t^{(i)}_{\mathrm{cyc}}(\omega) = t^{(i)}_{\mathrm{return}}(\omega) + \gamma\frac{2d^{(i)}_B}{v_{\mathrm{est}}(\omega)}, \qquad d^{(i)}_B = z_i - z_{\mathrm{bottom}}, \qquad \gamma = 0.95, \tag{S18}$$

The factor $\gamma = 0.95$ is a fixed margin in this timing prescription, not a reflection coefficient. For the closer monitor Σ_1, the estimates are

ω	1	50	100	150	200	300
$t^{(1)}_{\mathrm{cyc}}(\omega)$	13.58	7.30	6.02	6.34	5.06	3.21

(S19)

The smallest peak-based cycle estimate is approximately 3.21. These estimates provide approximate timing information, rather than bounds on the onset of returned probability. For the statistical decomposition, we use the common cutoff

$$t_{\mathrm{cut}} = 3.6. \tag{S20}$$

This cutoff defines a common numerical diagnostic window, not a strict return-free interval or an encounter-resolved decomposition of the wavefunction. A sensitivity check with an earlier cutoff is reported in Sec. S3.3.

S3.3. Early/late identity and reference values

For $T > t_{\mathrm{cut}}$, define

$$I_E = [0, t_{\mathrm{cut}}], \qquad I_L = (t_{\mathrm{cut}}, T]. \tag{S21}$$

The conditional weights are

$$w_L(T;\omega) = \int_{t_{\mathrm{cut}}}^{T} f_T(t;\omega)\, dt, \qquad w_E(T;\omega) = 1 - w_L(T;\omega). \tag{S22}$$

For positive early and late weights, as in all cases reported here, the corresponding conditional means are

$$t_E(\omega) = \frac{1}{w_E(T;\omega)} \int_0^{t_{\mathrm{cut}}} t f_T(t;\omega)\, dt, \tag{S23}$$

and

$$t_L(T;\omega) = \frac{1}{w_L(T;\omega)} \int_{t_{\mathrm{cut}}}^{T} t f_T(t;\omega)\, dt. \tag{S24}$$

They satisfy

$$\begin{aligned} \bar{\tau}_T^{\mathrm{cond}}(\omega) &= w_E(T;\omega) t_E(\omega) + w_L(T;\omega) t_L(T;\omega) \\ &= t_E(\omega) + w_L(T;\omega)\big[t_L(T;\omega) - t_E(\omega)\big]. \end{aligned} \tag{S25}$$

We define the late-window contribution[1] to the conditional mean,

$$\Delta t_L^{\mathrm{cond}}(T;\omega) := w_L(T;\omega)[t_L(T;\omega) - t_E(\omega)]. \tag{S26}$$

The absolute early detected probability is

$$D_E(\omega) := \int_0^{t_{\mathrm{cut}}} g(t;\omega)\, dt = D_T(\omega) w_E(T;\omega). \tag{S27}$$

Both D_E and t_E depend only on the prefix up to t_{cut}, and hence are unchanged by choosing $T = 10, 20, 30,$ or 40. The late quantities and the conditional weights do depend on T.

[1] For the six confinements in Eq. (S19), reducing the common cutoff to 3.21 preserves the decrease in early detected probability, the weak variation of its conditional mean, and the increase in both the late fraction among detections and the late conditional mean.

TABLE S2. Finite-window statistics at $T = 40$, with $t_{\rm cut} = 3.6$. The upper block contains spinor-ABC reference evolutions; the lower block contains componentwise scalar-ABC controls. Weights and window means refer to the corresponding conditional density f_{40}. Scalar statistics are reconstructed from normalized survival histories. Entries are rounded independently; displayed digits do not specify discretization accuracy.

ω	D_{40}	$\bar{\tau}_{40}^{\rm cond}$	$\mu^*(40)$	t_E	$w_L(40)$	$t_L(40)$	$\Delta t_L^{\rm cond}(40)$
Spinor ABC:							
1	0.952	4.663	6.358	2.595	0.394	7.839	2.067
25	0.916	5.980	8.832	2.561	0.405	10.999	3.419
50	0.887	7.158	10.878	2.543	0.447	12.864	4.615
75	0.862	8.020	12.433	2.533	0.479	13.982	5.488
100	0.841	8.683	13.666	2.526	0.504	14.735	6.157
150	0.808	9.598	15.436	2.517	0.538	15.677	7.081
200	0.784	10.178	16.618	2.511	0.559	16.234	7.666
300	0.740	11.113	18.631	2.501	0.590	17.088	8.613
Scalar ABC:							
1	0.956	4.372	5.923	2.598	0.386	7.198	1.774
300	0.956	4.375	5.926	2.600	0.386	7.195	1.775

Table S2 reports the reference values at $T = 40$. Across the sweep, D_E decreases from approximately 0.58 at $\omega = 1$ to 0.30 at $\omega = 300$, whereas t_E changes only from 2.59 to 2.50. The main prompt-sector effect is therefore a reduction in detected probability rather than a shift of its conditional mean to later times. At $T = 40$, the late fraction among detected events increases from 0.39 to 0.59, and the conditional mean within that sector increases from 7.84 to 17.09. This conditional fraction is distinct from the absolute late detection probability $D_L(T;\omega) = D_T(\omega)w_L(T;\omega)$.

Combining the conditional split with Eq. (S14) gives

$$\mu^*(T;\omega) = D_T(\omega)\big[t_E(\omega) + \Delta t_L^{\rm cond}(T;\omega)\big] + [1 - D_T(\omega)]T. \tag{S28}$$

Consequently the confinement dependence of the restricted mean cannot be identified with the delayed conditional contribution alone. These identities describe the signal; possible dynamical contributions from returns are discussed separately in Sec. S7.

Scalar-boundary control. The lower block of Table S2 removes the tangential coupling by imposing $\partial_z\Psi = i\kappa\Psi$ componentwise, retaining the bulk Hamiltonian, physical preparation, and reference geometry and spatial grid. The scalar rows are excluded from the spinor confinement fits.

Cutoff sensitivity. Repeating the decomposition at $T = 40$ with $t_{\rm cut} = 3.0$ for the same eight spinor-ABC reference evolutions preserves the suppression of early detected probability and the weak variation of its

conditional mean. Between $\omega = 1$ and 300, D_E decreases from 0.426 to 0.239, t_E changes from 2.353 to 2.295, and t_L increases from 6.532 to 15.328. The late fraction among detections increases overall, with a small initial decrease between $\omega = 1$ and 25; the absolute late probability is nonmonotonic for both cutoffs. The tabulated early/late statistics use $t_{\rm cut} = 3.6$.

S3.4. Observation-window comparison

Main-text Fig. 3 compares $T = 10, 20, 30, 40$ evaluated from the same eight evolutions. At $\omega = 300$, extending T from 20 to 40 adds detected probability $D_{40} - D_{20} \simeq 0.156$, corresponding to 37.4% of the probability surviving at $t = 20$. The survival contribution $TS(T)$ accounts for approximately 68.7% of $\mu^*(20)$ and 55.9% of $\mu^*(40)$. With $S(40) \simeq 0.260$, the unrestricted mean remains undetermined by the recorded histories.

For the eight spinor reference values in Table S2, the unweighted fit $\mu^*(40;\omega) \simeq 5.463 + 0.784\sqrt{\omega}$, reported in main-text Eq. (12), has a root-mean-square residual of 0.32 and a maximum absolute residual of 0.55, at $\omega = 25$. These residuals quantify agreement with the empirical fit.

S4. FIRST-PASS BULK INDEPENDENCE OF ω

We begin with an analytic bulk check. Before the packet has significantly touched the absorbing boundary, ordinary first-pass propagation cannot generate the observed confinement dependence on ω. In a separable, spin-independent bulk, changing ω only changes the normalized transverse factor; after cross-section integration, the leading Pauli flux is purely longitudinal and independent of ω.

We now show this explicitly. Let $z_b = L - \delta$ be an interior counting plane below the detector, and let

$$\Sigma_b = \{(x, y, z) : z = z_b\}. \tag{S29}$$

In the region below the detecting plane, assume the first-pass separable bulk Hamiltonian

$$H_{\rm bulk} = H_{xy}(\omega) + H_z, \tag{S30}$$

with

$$H_{xy}(\omega) = -\frac{1}{2}(\partial_x^2 + \partial_y^2) + V_{xy}(x, y; \omega), \qquad H_z = -\frac{1}{2}\partial_z^2. \tag{S31}$$

We also assume that the absorbing boundary has not yet significantly modified the state at the interior plane. For a prepared state with spatially constant spinor η, the auxiliary first-pass bulk field is

$$\Psi_{\rm fp}(x, y, z, t) = \chi_\omega(x, y, t)\, \phi_{\rm fp}(z, t)\, \eta, \qquad \rho_{\rm fp} = |\Psi_{\rm fp}|^2 = |\chi_\omega(x, y, t)|^2 |\phi_{\rm fp}(z, t)|^2. \tag{S32}$$

Here "first-pass" means the separable, spin-independent bulk evolution before appreciable feedback from the absorbing boundary. The longitudinal factor is initialized by $\phi_{\mathrm{fp}}(z,0)=\phi_0(z)$, and Ψ_{fp} is not the full spinor-ABC solution. It is used only to test whether ordinary bulk propagation can generate the observed ω-dependence. The spin density of this auxiliary field is

$$\boldsymbol{S}_{\mathrm{fp}}=\Psi_{\mathrm{fp}}^{\dagger}\boldsymbol{\sigma}\Psi_{\mathrm{fp}}=\rho_{\mathrm{fp}}\boldsymbol{s}_{\eta},\qquad \boldsymbol{s}_{\eta}=\eta^{\dagger}\boldsymbol{\sigma}\eta. \tag{S33}$$

The corresponding Pauli current is

$$\boldsymbol{j}_{\mathrm{fp}}^{P}=\boldsymbol{j}_{\mathrm{fp}}^{\mathrm{conv}}+\frac{1}{2}\boldsymbol{\nabla}\times\boldsymbol{S}_{\mathrm{fp}},\qquad \boldsymbol{j}_{\mathrm{fp}}^{\mathrm{conv}}=\mathrm{Im}(\Psi_{\mathrm{fp}}^{\dagger}\boldsymbol{\nabla}\Psi_{\mathrm{fp}}). \tag{S34}$$

Since $\boldsymbol{s}_{\eta}$ is constant in the first-pass separable approximation,

$$(\boldsymbol{\nabla}\times\boldsymbol{S}_{\mathrm{fp}})_z=\partial_x(\rho_{\mathrm{fp}}s_{\eta,\mathrm{y}})-\partial_y(\rho_{\mathrm{fp}}s_{\eta,\mathrm{x}}). \tag{S35}$$

Therefore

$$\int_{\Sigma_b}(\boldsymbol{\nabla}\times\boldsymbol{S}_{\mathrm{fp}})_z\,dx\,dy=\int_{\partial\Sigma_b}\left(\boldsymbol{S}_{\mathrm{fp},y}\nu_x-\boldsymbol{S}_{\mathrm{fp},x}\nu_y\right)d\ell=0, \tag{S36}$$

where $\boldsymbol{\nu}=(\nu_x,\nu_y)$ is the outward unit normal to $\partial\Sigma_b$ in the transverse plane. The last equality follows from the lateral boundary conditions in the finite box, or from transverse decay in the ideal waveguide. Hence the plane-integrated Pauli flux equals the plane-integrated convective flux:

$$\int_{\Sigma_b}j_{\mathrm{fp,z}}^{P}\,dx\,dy=\int_{\Sigma_b}j_{\mathrm{fp,z}}^{\mathrm{conv}}\,dx\,dy. \tag{S37}$$

Define the first-pass bulk flux through Σ_b by

$$\Phi_{\mathrm{bulk}}(t)=\int_{\Sigma_b}j_{\mathrm{fp,z}}^{P}(\boldsymbol{r},t)\,dA. \tag{S38}$$

Using Eq. (S32),

$$\begin{aligned}\Phi_{\mathrm{bulk}}(t)&=\int_{\Sigma_b}\mathrm{Im}(\Psi_{\mathrm{fp}}^{\dagger}\partial_z\Psi_{\mathrm{fp}})\,dA\\&=\left(\int|\chi_\omega(x,y,t)|^2\,dx\,dy\right)\mathrm{Im}\big(\phi_{\mathrm{fp}}^{*}\partial_z\phi_{\mathrm{fp}}\big)(z_b,t).\end{aligned} \tag{S39}$$

Since $H_{xy}(\omega)$ is self-adjoint with the chosen transverse boundary conditions, the transverse factor remains normalized. Therefore

$$\Phi_{\mathrm{bulk}}(t)=\mathrm{Im}\big(\phi_{\mathrm{fp}}^{*}\partial_z\phi_{\mathrm{fp}}\big)(z_b,t), \tag{S40}$$

which is independent of ω and independent of the detector parameter κ. When the integrals converge absolutely and the denominator is nonzero, the corresponding flux-weighted time is

$$\bar{\tau}_{\mathrm{bulk}}=\frac{\int_0^\infty t\,\Phi_{\mathrm{bulk}}(t)\,dt}{\int_0^\infty\Phi_{\mathrm{bulk}}(t)\,dt} \tag{S41}$$

which is ω-independent under the stated assumptions. This does not claim that the full finite-box detector signal is ω-independent. It only shows that the leading first-pass bulk transport does not generate the confinement trend; the effect must enter through the spinor absorbing boundary.

S5. BOUNDARY-SYMBOL MECHANISM

This section studies the detailed boundary-symbol calculation used in the main text. The calculation is a local probe of the spinor ABC, not a second absorber and not a physical layer below the detector. The physical detector observable remains the detector flux at $z = L$. At the detecting plane the spinor ABC is

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\Psi = i\kappa \boldsymbol{\sigma}_z \Psi, \qquad \Psi = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}. \tag{S42}$$

In components,

$$\begin{aligned} \partial_z \psi_\uparrow &= i\kappa \psi_\uparrow - (\partial_x - i\partial_y)\psi_\downarrow, \\ \partial_z \psi_\downarrow &= i\kappa \psi_\downarrow + (\partial_x + i\partial_y)\psi_\uparrow. \end{aligned} \tag{S43}$$

Equivalently, the boundary condition is the first-order normal boundary relation

$$\partial_z \Psi\big|_{z=L} = \mathcal{C}\Psi\big|_{z=L}, \qquad \mathcal{C} = \begin{pmatrix} i\kappa & -D_- \\ D_+ & i\kappa \end{pmatrix}, \qquad D_\pm = \partial_x \pm i\partial_y. \tag{S44}$$

Therefore the detector does not impose only a scalar normal impedance. It couples the normal boundary relation to tangential derivatives of the boundary trace.

S5.1. Exact near-boundary identity

We now isolate the local normal response generated by the boundary relation itself. Fix a time t and take the boundary trace $\Psi(\cdot, L, t)$ as given. We use the homogeneous relation $\partial_z \Psi = \mathcal{C}\Psi$ only to continue this trace a small auxiliary distance inward, to $z = L - \epsilon$ with $\epsilon > 0$. This continuation is a local probe of the boundary symbol, not a physical detector layer; the true time dependent Schrödinger equation (TDSE) solution need not satisfy $\partial_z \Psi = \mathcal{C}\Psi$ away from the detecting plane. Define the boundary-condition defect/error in a small interval $L - \delta < z \le L$ by

$$\mathcal{E}_{\mathrm{bc}}(z, t) := \partial_z \Psi(\cdot, z, t) - \mathcal{C}\Psi(\cdot, z, t). \tag{S45}$$

The spinor ABC implies

$$\mathcal{E}_{\mathrm{bc}}(L, t) = 0. \tag{S46}$$

After tangential Fourier transform, each mode satisfies

$$\partial_z \widehat{\Psi}(\boldsymbol{\xi}, z, t) = \mathcal{C}(\boldsymbol{\xi})\widehat{\Psi}(\boldsymbol{\xi}, z, t) + \widehat{\mathcal{E}}_{\mathrm{bc}}(\boldsymbol{\xi}, z, t). \tag{S47}$$

Variation of constants, integrated backward from L to z, gives the exact near-detector identity

$$\widehat{\Psi}(\boldsymbol{\xi}, z, t) = e^{(z-L)\mathcal{C}(\boldsymbol{\xi})}\widehat{\Psi}(\boldsymbol{\xi}, L, t) - \int_z^L e^{(z-s)\mathcal{C}(\boldsymbol{\xi})}\widehat{\mathcal{E}}_{\mathrm{bc}}(\boldsymbol{\xi}, s, t)\, ds. \tag{S48}$$

The first term is the homogeneous normal continuation generated by the spinor ABC. The second term is the Duhamel remainder. We denote the boundary trace by

$$W_\omega(\boldsymbol{\xi}, t) := \widehat{\Psi}_\omega(\boldsymbol{\xi}, L, t). \tag{S49}$$

For $z = L - \epsilon$, the Duhamel remainder is

$$\widehat{\mathcal{E}}_{\mathrm{Duh}}(\boldsymbol{\xi}, L-\epsilon, t) := -\int_0^\epsilon e^{-(\epsilon-u)\mathcal{C}(\boldsymbol{\xi})} \widehat{\mathcal{E}}_{\mathrm{bc}}(\boldsymbol{\xi}, L-u, t)\, du. \tag{S50}$$

This is the same remainder term denoted by $\widehat{\mathcal{R}}_{\mathrm{Duh}}$ in the main text. Under the regularity assumption quantified in Sec. S6.1, this remainder is $O(\epsilon^2)$ modewise as $\epsilon \to 0$. Hence it has no zeroth- or first-order contribution at the detecting plane. The leading inward normal response is therefore determined by the homogeneous boundary symbol.

S5.2. Tangential symbol and spin–momentum branches

The tangential Fourier transform is used only to expose the local symbol of the boundary operator. In the finite-difference calculation, the analogous object is the discrete tangential symbol used in Sec. S6. For a tangential Fourier mode

$$\boldsymbol{\xi} = (\xi_x, \xi_y), \qquad \xi_\pm = \xi_x \pm i\xi_y, \tag{S51}$$

Eq. (S44) gives

$$\mathcal{C}(\boldsymbol{\xi}) = \begin{pmatrix} i\kappa & -i\xi_- \\ i\xi_+ & i\kappa \end{pmatrix}. \tag{S52}$$

Separate the scalar impedance from the tangential spin coupling:

$$\mathcal{J}(\boldsymbol{\xi}) := \mathcal{C}(\boldsymbol{\xi}) - i\kappa I. \tag{S53}$$

Then

$$\mathcal{J}(\boldsymbol{\xi}) = (\hat{\boldsymbol{z}} \times \boldsymbol{\xi}) \cdot \boldsymbol{\sigma} = -\xi_y \sigma_x + \xi_x \sigma_y, \qquad \mathcal{J}(\boldsymbol{\xi})^2 = R^2 I, \qquad R = |\boldsymbol{\xi}|. \tag{S54}$$

For $R > 0$, define

$$\Gamma(\boldsymbol{\xi}) = \frac{\mathcal{J}(\boldsymbol{\xi})}{R} = (\hat{\boldsymbol{z}} \times \hat{\boldsymbol{\xi}}) \cdot \boldsymbol{\sigma}, \qquad \Gamma(\boldsymbol{\xi})^2 = I. \tag{S55}$$

The zero mode $R = 0$ is degenerate and carries no $R\Gamma$ contribution; it is understood by continuity. Therefore

$$\mathcal{C}(\boldsymbol{\xi}) = i\kappa I + R\Gamma(\boldsymbol{\xi}). \tag{S56}$$

The two boundary eigenbranches are

$$\lambda_\pm(\boldsymbol{\xi}) = i\kappa \pm R, \tag{S57}$$

with projectors

$$\Pi_{\pm}(\boldsymbol{\xi}) = \frac{1}{2}\big(I \pm \Gamma(\boldsymbol{\xi})\big). \tag{S58}$$

The projectors select two tangential spin–momentum branches, with in-plane spin direction perpendicular to the tangential momentum. This is the key local difference between the spinor ABC and a scalar absorbing boundary.

S5.3. Auxiliary normal response

Since $\Gamma^2 = I$, the homogeneous exponential in Eq. (S48) is explicit:[2]

$$\begin{aligned} e^{-\epsilon\mathcal{C}(\boldsymbol{\xi})} &= e^{-i\kappa\epsilon} e^{-\epsilon R\Gamma(\boldsymbol{\xi})} \\ &= e^{-i\kappa\epsilon}\left[e^{-R\epsilon}\Pi_{+}(\boldsymbol{\xi}) + e^{R\epsilon}\Pi_{-}(\boldsymbol{\xi})\right]. \end{aligned} \tag{S59}$$

Putting $z = L - \epsilon$ in Eq. (S48) gives

$$\widehat{\Psi}_{\omega}(\boldsymbol{\xi}, L-\epsilon, t) = e^{-i\kappa\epsilon} B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W_{\omega}(\boldsymbol{\xi}, t) + \widehat{\mathcal{R}}_{\mathrm{Duh}}(\boldsymbol{\xi}, L-\epsilon, t), \tag{S60}$$

where the two-branch normal-response map is

$$B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) = e^{-R\epsilon}\Pi_{+}(\boldsymbol{\xi}) + e^{R\epsilon}\Pi_{-}(\boldsymbol{\xi}). \tag{S61}$$

This is the matrix-valued branch filter acting on spinor amplitudes. The factor $e^{-R\epsilon}\Pi_{+}$ is the boundary-induced evanescent normal factor, with length scale R^{-1}, and the term $e^{R\epsilon}\Pi_{-}$ is the companion branch of the same auxiliary inward continuation. It is not physical amplification, not an absorbed component, and not a second detector model. Using Eq. (S58),

$$B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) = \cosh(R\epsilon) I - \sinh(R\epsilon)\Gamma(\boldsymbol{\xi}) = e^{-R\epsilon\Gamma(\boldsymbol{\xi})}. \tag{S62}$$

Thus the branch structure is not visible as a multiplicative suppression of the detector density itself, since $B_{\mathrm{br}}(0, \boldsymbol{\xi}) = I$. It enters through the first inward normal response:

$$\partial_{\epsilon} B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi})|_{\epsilon=0} = -R\Gamma(\boldsymbol{\xi}) = -\mathcal{J}(\boldsymbol{\xi}). \tag{S63}$$

Equation (S63) shows that R sets the scale of the first inward normal response for each tangential Fourier component. For the incident transverse ground-state family, $\ell_{\perp} = \omega^{-1/2}$, giving the characteristic preparation scale[3]

$$R = |\boldsymbol{\xi}| \sim \ell_{\perp}^{-1} \sim \sqrt{\omega}. \tag{S64}$$

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[2] For $R > 0$, $C(\boldsymbol{\xi})$ is diagonalizable with eigenvalues $\lambda_{\pm}(\boldsymbol{\xi}) = i\kappa \pm R$ and spectral projectors $\Pi_{\pm}(\boldsymbol{\xi})$. The Lagrange–Sylvester functional calculus gives $f(C(\boldsymbol{\xi})) = f(\lambda_{+})\Pi_{+}(\boldsymbol{\xi}) + f(\lambda_{-})\Pi_{-}(\boldsymbol{\xi})$ for analytic f. Taking $f(\lambda) = e^{(z-L)\lambda}$, and then $z = L - \epsilon$, gives Eq. (S59). The degenerate $R = 0$ mode is obtained by continuity.

[3] Equivalently, for normalized transverse oscillator modes centered at the trap center, $D_{\pm}\chi_{00}^{(\omega)} = -\sqrt{\omega}\,\chi_{\pm}^{(\omega)}$, $\chi_{\pm}^{(\omega)} = \frac{\chi_{10}^{(\omega)} \pm i\chi_{01}^{(\omega)}}{\sqrt{2}}$, up to phase conventions. This is the oscillator-basis version of the same boundary-symbol scale $R = |\boldsymbol{\xi}| \sim \sqrt{\omega}$.

The evolved boundary trace $W_\omega(\boldsymbol{\xi}, t)$ is determined by the detector-present dynamics and need not retain this incident momentum distribution.

Connection with local scattering. The same projectors diagonalize the frozen local matching problem (satisfying $\Pi_\pm \Psi = \Psi$) in main-text Sec. 4(ii). On a fixed tangential branch, the boundary relation at $z = L$ becomes

$$\partial_z \psi_\pm = (i\kappa \pm R)\psi_\pm. \tag{S65}$$

Therefore even the local no-click matching problem is governed by the non-scalar pair $i\kappa \pm R$, not by a single scalar Robin impedance. In a frozen half-space plane-wave check this corresponds to a matrix no-click reflection amplitude of the form

$$\mathrm{Ref}_{\mathrm{loc}}(k_n, \boldsymbol{\xi}) = r_+(k_n, R)\Pi_+(\boldsymbol{\xi}) + r_-(k_n, R)\Pi_-(\boldsymbol{\xi}), \tag{S66}$$

Here $k_n > 0$, and the amplitudes $r_\pm$ are given in main-text Sec. 4(ii). Since $|r_\pm|^2 = 1 - A$, we write $r_\pm = \sqrt{1-A}\, e^{i\phi_\pm}$, where $\phi_\pm = \arg r_\pm$ for $R > 0$. We then obtain

$$\mathrm{Ref}_{\mathrm{loc}}^\dagger \mathrm{Ref}_{\mathrm{loc}} = (1-A)I, \qquad \mathrm{Ref}_{\mathrm{loc}} = \sqrt{1-A}\, e^{i\phi_0} \exp\!\left(\frac{i\Delta\phi}{2}\Gamma(\boldsymbol{\xi})\right), \tag{S67}$$

where $A = A(k_n, R)$, $\phi_\pm = \arg r_\pm(k_n, R)$, $\phi_0 = (\phi_+ + \phi_-)/2$, and $\Delta\phi = \phi_+ - \phi_-$. Thus, at fixed $(k_n, \boldsymbol{\xi})$, reflection produces equal attenuation in both branches and a unitary spin rotation about $\mathbf{n}_\xi$, acting nontrivially on coherent branch superpositions. The reflection operator maps incident to reflected amplitudes, whereas B_{br} continues a prescribed boundary trace inward; neither replaces the complete confined-wavepacket evolution.

S5.4. Covariance form of the local boundary response

The previous subsection identifies the local boundary scale. We now express its first-order effect on a normalized auxiliary time moment in covariance form. This auxiliary moment is only a diagnostic of the local boundary response; the physical detector statistic remains the detector flux at $z = L$.

Since the scalar phase $e^{-i\kappa\epsilon}$ drops out of densities, the relevant first-order density response is obtained after removing this phase. By Eq. (S60) and the Duhamel estimate of Sec. S6.1,

$$e^{i\kappa\epsilon}\widehat{\Psi}_\omega(\boldsymbol{\xi}, L-\epsilon, t) = B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W_\omega(\boldsymbol{\xi}, t) + O_{\mathrm{Duh}}(\epsilon^2). \tag{S68}$$

Thus the exact phase-removed trace and the homogeneous branch probe have the same zeroth and first inward variations at the detecting plane. We therefore define

$$\widetilde{W}_{\omega,\epsilon}(\boldsymbol{\xi}, t) := B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) W_\omega(\boldsymbol{\xi}, t), \tag{S69}$$

and compute the first normal variation from this homogeneous branch probe. We keep the branch exponential exact, rather than replacing it by a finite-depth Taylor approximation. Since $B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) = e^{-\epsilon R \Gamma(\boldsymbol{\xi})}$ and $\Gamma(\boldsymbol{\xi})$ is Hermitian with $\Gamma^2 = I$, we have

$$B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi})^\dagger B_{\mathrm{br}}(\epsilon, \boldsymbol{\xi}) = e^{-2\epsilon R \Gamma(\boldsymbol{\xi})} = \cosh(2R\epsilon) I - \sinh(2R\epsilon)\Gamma(\boldsymbol{\xi}). \tag{S70}$$

Therefore the auxiliary homogeneous boundary-layer spectral density is exactly

$$\rho^{\mathrm{bl}}_{\omega,\epsilon}(\boldsymbol{\xi}, t) := \left| \widetilde{W}_{\omega,\epsilon}(\boldsymbol{\xi}, t) \right|^2 = W_\omega^\dagger e^{-2\epsilon R \Gamma(\boldsymbol{\xi})} W_\omega. \tag{S71}$$

Where $W^\dagger W = 0$, a_ω may be defined arbitrarily, e.g. $a_\omega = 0$. Writing

$$\rho^{\mathrm{bl}}_{\omega,0}(\boldsymbol{\xi}, t) = W_\omega^\dagger W_\omega, \quad \text{and} \qquad a_\omega(\boldsymbol{\xi}, t) := -2 \frac{W_\omega^\dagger \Gamma(\boldsymbol{\xi}) W_\omega}{W_\omega^\dagger W_\omega}, \tag{S72}$$

we obtain the exact reweighting form

$$\rho^{\mathrm{bl}}_{\omega,\epsilon}(\boldsymbol{\xi}, t) = \rho^{\mathrm{bl}}_{\omega,0}(\boldsymbol{\xi}, t)\, b_{\omega,\epsilon}(\boldsymbol{\xi}, t), \tag{S73}$$

where

$$b_{\omega,\epsilon}(\boldsymbol{\xi}, t) = \cosh(2R\epsilon) + \frac{a_\omega(\boldsymbol{\xi}, t)}{2} \sinh(2R\epsilon). \tag{S74}$$

The quantity a_ω is real because Γ is Hermitian. Since the eigenvalues of Γ are ± 1,

$$|a_\omega(\boldsymbol{\xi}, t)| \le 2. \tag{S75}$$

In particular,

$$b_{\omega,0} = 1, \qquad \partial_\epsilon b_{\omega,\epsilon}|_{\epsilon=0} = R a_\omega(\boldsymbol{\xi}, t). \tag{S76}$$

No finite-depth smallness assumption $R\epsilon \ll 1$ is used here: the branch map has been kept exact, and only its derivative at the physical detecting plane is taken. We now pass from this pointwise density reweighting to a normalized auxiliary time moment. Let

$$d\mu^{(0)}_\omega(\boldsymbol{\xi}, t) := \rho^{\mathrm{bl}}_{\omega,0}(\boldsymbol{\xi}, t)\, d^2\boldsymbol{\xi}\, dt, \qquad Z^{(0)}_\omega := \int d\mu^{(0)}_\omega, \tag{S77}$$

where the time integral is over the chosen finite observation window. Define the normalized boundary-trace measure

$$d\nu^{(0)}_\omega := \frac{d\mu^{(0)}_\omega}{Z^{(0)}_\omega}. \tag{S78}$$

For $D_T > 0$, constant κ, and the same observation window, the temporal marginal of $\nu^{(0)}_\omega$ is $g(t;\omega)/D_T(\omega)$. The branch-reweighted auxiliary measure at depth ϵ is

$$d\mu^{\mathrm{bl}}_{\omega,\epsilon} = \rho^{\mathrm{bl}}_{\omega,\epsilon}(\boldsymbol{\xi}, t)\, d^2\xi\, dt = b_{\omega,\epsilon}(\boldsymbol{\xi}, t)\, d\mu^{(0)}_\omega. \tag{S79}$$

After normalization,

$$d\nu^{\mathrm{bl}}_{\omega,\epsilon} = \frac{d\mu^{\mathrm{bl}}_{\omega,\epsilon}}{\int d\mu^{\mathrm{bl}}_{\omega,\epsilon}} = \frac{b_{\omega,\epsilon}(\boldsymbol{\xi},t)}{\mathbb{E}_{\nu^{(0)}_{\omega}}[b_{\omega,\epsilon}]}\, d\nu^{(0)}_{\omega}. \tag{S80}$$

This identity is exact for the homogeneous branch probe. Let $F(\boldsymbol{\xi},t)$ be real and independent of ϵ. Assume $0 < Z^{(0)}_{\omega} < \infty$, that the following expectations are finite, and that the auxiliary measures are normalizable near $\epsilon = 0^{+}$, with differentiation permitted under the integrals. Then

$$\mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[F] = \frac{\mathbb{E}_{\nu^{(0)}_{\omega}}[F\, b_{\omega,\epsilon}]}{\mathbb{E}_{\nu^{(0)}_{\omega}}[b_{\omega,\epsilon}]}. \tag{S81}$$

Differentiating Eq. (S81) at $\epsilon = 0$, and using Eq. (S76) gives

$$\partial_{\epsilon}\mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[F]\Big|_{\epsilon=0} = \mathbb{E}_{\nu^{(0)}_{\omega}}[F\, Ra_{\omega}] - \mathbb{E}_{\nu^{(0)}_{\omega}}[F]\, \mathbb{E}_{\nu^{(0)}_{\omega}}[Ra_{\omega}]. \tag{S82}$$

Thus

$$\partial_{\epsilon}\mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[F]\Big|_{\epsilon=0} = \mathrm{Cov}_{\nu^{(0)}_{\omega}}(F, Ra_{\omega})\,. \tag{S83}$$

Taking $F = t$ to calculate the normalized first time moment, the local first-order boundary-response coefficient is

$$\Lambda_{\omega} := \partial_{\epsilon}\bar{t}_{\mathrm{bl}}(\omega,\epsilon)|_{\epsilon=0}\,, \qquad \bar{t}_{\mathrm{bl}}(\omega,\epsilon) := \mathbb{E}_{\nu^{\mathrm{bl}}_{\omega,\epsilon}}[t], \tag{S84}$$

and therefore

$$\Lambda_{\omega} = \mathrm{Cov}_{\nu^{(0)}_{\omega}}(t, Ra_{\omega})\,. \tag{S85}$$

This is the desired covariance form of the local boundary response. The derivation used the exact branch exponential through $b_{\omega,\epsilon}$, and the only derivative taken was the derivative at the physical detecting plane, $\epsilon = 0$. For the harmonic transverse family we write

$$R = \sqrt{\omega}\, s, \tag{S86}$$

where s is the dimensionless tangential radius in oscillator units. Hence

$$\Lambda_{\omega} = \sqrt{\omega}\ \mathrm{Cov}_{\nu^{(0)}_{\omega}}(t, sa_{\omega}) = \sqrt{\omega}\,\beta^{\mathrm{bl}}_{\omega}, \tag{S87}$$

with

$$\beta^{\mathrm{bl}}_{\omega} := \mathrm{Cov}_{\nu^{(0)}_{\omega}}(t, sa_{\omega})\,. \tag{S88}$$

Since $|a_{\omega}| \le 2$ and $0 \le t \le T$ on the finite observation window, Cauchy–Schwarz gives, for example,

$$|\beta^{\mathrm{bl}}_{\omega}| \le 2T\sqrt{\mathbb{E}_{\nu^{(0)}_{\omega}}[s^2]}. \tag{S89}$$

At fixed T, Eq. (S89) bounds $\beta^{\mathrm{bl}}_{\omega}$ uniformly on any family for which the scaled second moment is uniformly bounded. Table S3 reports the moments and covariances at the sampled confinements; a

bound on the magnitude does not fix the sign or variation of the coefficient. For $D_T > 0$, constant κ, and the same observation window, the temporal marginal of $\nu_\omega^{(0)}$ is $g(t;\omega)/D_T(\omega)$, so $\bar{t}^{\mathrm{bl}}(\omega,0) = \bar{\tau}_T^{\mathrm{cond}}(\omega)$. The derivative Λ_ω measures the initial change of this normalized moment under the prescribed inward branch reweighting. Its sign is set by the correlation of time with the spin-resolved tangential factor Ra_ω. Equation (S87) separates the oscillator wave-number scale from this trace-dependent correlation. Confinement-dependent changes of the physical detection law are obtained by comparing the corresponding detector-present evolutions.

S6. FINITE-GRID BOUNDARY-SYMBOL DIAGNOSTIC AND DUHAMEL ESTIMATE

We use the eight reference evolutions of Secs. S2–S3 over $0 \le t \le 40$ to evaluate the boundary-weighted covariance, check the recorded detector flux, and compare the auxiliary inward continuation with the neighbouring numerical plane. Section S6.1 gives the continuum remainder estimate; Sec. S6.2 examines the spin and branch composition of the detector signal. Section S6.3 reports a separate $T = 40$ refinement series at $\omega = 300$.

Boundary trace and discrete symbol. The auxiliary probe depth is one recorded longitudinal spacing,

$$\epsilon_g = h_z, \tag{S90}$$

with $h_z \simeq 0.013$. The boundary trace is evaluated at the numerical detecting plane z_D of Table S1, and the inward trace at the preceding stored plane z_{N_z-2}. The continuum estimate in Sec. S6.1 retains the boundary coordinate L. We use the notation of Sec. S5 for the numerical boundary trace,

$$W_\omega(\boldsymbol{\xi}, t) = \widehat{\Psi}_\omega(\boldsymbol{\xi}, z_D, t), \tag{S91}$$

where $\boldsymbol{\xi} = (\xi_x, \xi_y)$ labels the orthonormal tangential Fourier transform. The central-difference symbol is

$$\boldsymbol{\xi}_h(\boldsymbol{\xi}) = \left(\frac{\sin(\xi_x h_x)}{h_x}, \frac{\sin(\xi_y h_y)}{h_y} \right), \qquad R_h = |\boldsymbol{\xi}_h|, \qquad s = \frac{R_h}{\sqrt{\omega}}. \tag{S92}$$

Set $J_h = (\widehat{\mathbf{z}} \times \boldsymbol{\xi}_h) \cdot \boldsymbol{\sigma}$. Then $J_h^2 = R_h^2 I$ and

$$\widehat{J}_h = \frac{J_h}{R_h} \qquad (R_h > 0). \tag{S93}$$

This is the local stencil symbol, not an exact diagonalization of the finite transverse Dirichlet problem. Numerically, we treat $R_h \le r_{0,h}$ as neutral, with $r_{0,h} = 64\epsilon_{\mathrm{mach}} \max(1, \max R_h)$ in double precision. The even 100×100 Fourier grid has four zero/Nyquist combinations. On nonneutral modes with nonzero density, the spin factor is

$$a_\omega(\boldsymbol{\xi}, t) = -2 \frac{W_\omega^\dagger \widehat{J}_h W_\omega}{W_\omega^\dagger W_\omega}, \qquad |a_\omega| \le 2. \tag{S94}$$

We set $a_\omega = 0$ on neutral or zero-density modes. Neutral modes remain in the total boundary weight but contribute neither to the $R_h a_\omega$ nor to the $s a_\omega$ numerator.

Boundary-weighted moments. For a horizon T with nonzero integrated boundary density, define

$$\mathbb{E}_{\nu_\omega^{(0)}}[F] = \frac{\displaystyle\int_0^T \sum_{\boldsymbol{\xi}} \|W_\omega(\boldsymbol{\xi}, t)\|^2 F(\boldsymbol{\xi}, t)\, dt}{\displaystyle\int_0^T \sum_{\boldsymbol{\xi}} \|W_\omega(\boldsymbol{\xi}, t)\|^2\, dt}. \tag{S95}$$

The transverse quadrature factor $h_x h_y$ cancels in this ratio. The recorded density and moment numerators are interpolated linearly on their actual diagnostic times. Both their integrals and first time moments are evaluated from these interpolants. The finite-grid counterparts of the coefficients in Sec. S5 are

$$\beta_\omega^{\mathrm{bl}} = \mathrm{Cov}_{\nu_\omega^{(0)}}(t, s a_\omega), \qquad \Lambda_\omega = \mathrm{Cov}_{\nu_\omega^{(0)}}(t, R_h a_\omega) = \sqrt{\omega}\, \beta_\omega^{\mathrm{bl}}. \tag{S96}$$

Their dependence on T is implicit. The dimensionless probe scale

$$q_\omega = \mathbb{E}_{\nu_\omega^{(0)}}[(h_z R_h)^2] = (h_z \sqrt{\omega})^2 \mathbb{E}_{\nu_\omega^{(0)}}[s^2] \tag{S97}$$

quantifies the sampled tangential scale at this depth; it is not an error estimate or a smallness assumption in the covariance derivation.

Flux consistency. Let $P(t)$ be the stored squared norm of Sec. S2, $P_0 = P(0)$, and $N_D(t) = h_x h_y \sum_{\boldsymbol{\xi}} \|W_\omega(\boldsymbol{\xi}, t)\|^2$ the transverse squared-norm integral at the detecting plane. The endpoint-density budget check is

$$\Delta_{\mathrm{bud}}^{\mathrm{rel}} = \frac{\left| \displaystyle\int_0^T \kappa N_D(t)\, dt - [P_0 - P(T)] \right|}{P_0 - P(T)}. \tag{S98}$$

The boundary-density integral is evaluated by trapezoidal quadrature on the native plane-record times, separately from the accepted-step Crank–Nicolson (CN) event sum of Sec. S3.

We also compare the normalized boundary-density rate $J_{\rho,h} = \kappa N_D / P_0$ with the accepted-step detector rate g_{CN}. On the native plane-record intervals I_j, of duration Δ_j, partitioning $[0, 40]$, define

$$e(f, g) = \frac{1}{D_{40}} \sum_j \Delta_j |\overline{f}_j - \overline{g}_j|. \tag{S99}$$

Here D_{40} is from the same run. Bars denote interval-averaged rates: CN event probabilities are accumulated with interval overlap and divided by Δ_j; endpoint rates are trapezoid-averaged. The absolute values prevent cancellation between intervals. For the $\omega = 300$ reference evolution, the interval-averaged discrepancy is $e(J_{\rho,h}, g_{\mathrm{CN}}) \simeq 9.43 \times 10^{-4}$.

For the unrounded semidiscrete last-row stencil, the normalized current entering the final numerical cell satisfies

$$J_{\mathrm{link},h} = J_{\rho,h} + \frac{dP_{\mathrm{cell}}}{dt}, \qquad P_{\mathrm{cell}} = \frac{h_z N_D}{2P_0}. \tag{S100}$$

Thus current entering that cell includes the change of its stored probability as well as permanent detector loss. At $\omega = 300$, the CN midpoint link current, integrated over the transverse grid and both spin components, has discrepancy 5.29×10^{-3} from g_{CN} on the native plane-record intervals. Subtracting the normalized terminal-cell probability change divided by the interval duration reduces this discrepancy to 5.64×10^{-7}. This is an interval-averaged consistency check of the discrete probability balance within the same evolution.

One-grid inward continuation. We compare the homogeneous continuation generated by the boundary symbol with the recorded field on the preceding longitudinal plane:

$$\begin{aligned} W_{\mathrm{hom}}(\boldsymbol{\xi}, t; h_z) &= e^{-i\kappa h_z} e^{-h_z J_h} W_\omega(\boldsymbol{\xi}, t), \\ W_{\mathrm{in}}(\boldsymbol{\xi}, t) &= \widehat{\Psi}_\omega(\boldsymbol{\xi}, z_{N_z-2}, t). \end{aligned} \tag{S101}$$

The reported discrepancy is

$$\mathcal{E}_{2\mathrm{br}}(h_z) = \left\langle \frac{\|W_{\mathrm{in}}(\cdot, t) - W_{\mathrm{hom}}(\cdot, t; h_z)\|_{\ell^2_\xi}}{\|W_{\mathrm{in}}(\cdot, t)\|_{\ell^2_\xi}} \right\rangle_{\mathrm{valid}}. \tag{S102}$$

Here $\langle f \rangle_{\mathrm{valid}} = \int_V N_D(t) f(t)\, dt / \int_V N_D(t)\, dt$, where V consists of diagnostic intervals with finite, valid discrepancy values at both endpoints. We integrate the product of the separate linear density and discrepancy interpolants. Excluded intervals enter neither integral, and no additional signal-amplitude threshold is applied.

TABLE S3. Finite-grid diagnostics at $T = 40$. The endpoint budget uses the native plane-record times; the moments and continuation discrepancy use the native spin/branch times. The valid-interval average is defined in Eq. (S102). Displayed digits do not specify discretization accuracy.

ω	$h_z\sqrt{\omega}$	q_ω	$\Delta^{\mathrm{rel}}_{\mathrm{bud}}$	$\mathbb{E}[s^2]$	$\beta^{\mathrm{bl}}_\omega$	Λ_ω	$\mathcal{E}_{2\mathrm{br}}$
1	0.013	1.75×10^{-4}	1.81×10^{-5}	0.982	-0.16	-0.16	8.90×10^{-4}
25	0.067	8.74×10^{-3}	1.61×10^{-4}	1.966	-0.255	-1.274	3.47×10^{-3}
50	0.094	2.44×10^{-2}	5.15×10^{-4}	2.745	-0.060	-0.424	6.55×10^{-3}
75	0.115	4.44×10^{-2}	2.67×10^{-4}	3.333	-0.015	-0.134	9.78×10^{-3}
100	0.133	6.77×10^{-2}	7.13×10^{-4}	3.808	0.001	0.012	1.32×10^{-2}
150	0.163	1.21×10^{-1}	3.84×10^{-4}	4.522	0.008	0.099	2.07×10^{-2}
200	0.189	1.78×10^{-1}	6.69×10^{-4}	5.009	0.004	0.053	2.87×10^{-2}
300	0.231	2.95×10^{-1}	9.43×10^{-4}	5.524	0.001	0.025	4.56×10^{-2}

Table S3 gives $0.982 \le \mathbb{E}[s^2] \le 5.524$. On the reference grid, the covariance $\beta^{\mathrm{bl}}_\omega$ is negative at $\omega = 1, 25, 50, 75$ and positive at $\omega = 100, 150, 200, 300$. The identity $\Lambda_\omega = \sqrt{\omega}\, \beta^{\mathrm{bl}}_\omega$ separates the oscillator scale from a trace-dependent correlation; it does not determine the restricted-mean fit coefficient. The

largest tabulated endpoint-budget mismatch is 9.43×10^{-4}, and the largest averaged continuation discrepancy is 4.56%.

S6.1. Duhamel estimate

We now give the continuum estimate underlying the auxiliary first variation. For fixed $(\boldsymbol{\xi}, t)$, assume that the boundary-condition defect $\widehat{E}_{\mathrm{bc}}$ of Sec. S5.1 is C^1 in a normal neighbourhood of L, and define

$$\mathcal{G}_\omega(\boldsymbol{\xi}, t; \epsilon) = \sup_{0 \le u \le \epsilon} \|\partial_z \widehat{E}_{\mathrm{bc}}(\boldsymbol{\xi}, L-u, t)\|. \tag{S103}$$

Since $\widehat{E}_{\mathrm{bc}}(\boldsymbol{\xi}, L, t) = 0$,

$$\|\widehat{E}_{\mathrm{bc}}(\boldsymbol{\xi}, L-u, t)\| \le u \mathcal{G}_\omega(\boldsymbol{\xi}, t; \epsilon). \tag{S104}$$

The Hermitian part of $C(\boldsymbol{\xi})$ has eigenvalues $\pm R$, so

$$\|e^{-(\epsilon-u)C(\boldsymbol{\xi})}\| \le e^{(\epsilon-u)R} \le e^{\epsilon R}. \tag{S105}$$

Inserting these bounds into the Duhamel remainder of Sec. S5.1 gives

$$\begin{aligned} \|\widehat{E}_{\mathrm{Duh}}(\boldsymbol{\xi}, L-\epsilon, t)\| &\le e^{\epsilon R} \mathcal{G}_\omega(\boldsymbol{\xi}, t; \epsilon) \int_0^\epsilon u \, du \\ &= \frac{\epsilon^2}{2} e^{\epsilon R} \mathcal{G}_\omega(\boldsymbol{\xi}, t; \epsilon). \end{aligned} \tag{S106}$$

Local boundedness of G_ω therefore yields, modewise,

$$\widehat{E}_{\mathrm{Duh}}(\boldsymbol{\xi}, L-\epsilon, t) = O(\epsilon^2) \qquad (\epsilon \to 0^+), \tag{S107}$$

and

$$\widehat{E}_{\mathrm{Duh}}(\boldsymbol{\xi}, L, t) = 0, \qquad \partial_\epsilon \widehat{E}_{\mathrm{Duh}}(\boldsymbol{\xi}, L-\epsilon, t)\Big|_{\epsilon=0^+} = 0. \tag{S108}$$

The remainder thus leaves the first inward normal variation unchanged. This is a modewise estimate; no bound uniform in $\boldsymbol{\xi}$ or ω is asserted. The one-depth comparison in Table S3 does not test the asymptotic ϵ^2 order.

S6.2. Boundary branches and particle-spin polarization

For the same eight evolutions, initialized with $\eta = (1, 0)^{\mathsf{T}}$, we compare two distinct quantities: the tangential branch composition at the detecting plane and the particle-spin polarization associated with detection or survival. Table S4 summarizes all eight confinements; Fig. S1 shows the time dependence for $\omega = 1$ and 300.

Tangential branch composition. For $R_h > r_{0,h}$, the finite-difference counterparts of the projectors in Eq. (S58) (main-text Eq. (22)) are

$$\Pi_{\pm,h} = \frac{1}{2}\left(I \pm \frac{J_h}{R_h}\right). \tag{S109}$$

They act separately on each Fourier component of the evolved boundary trace. With the orthonormal Fourier convention, define

$$N_\pm(t) = h_x h_y \sum_{R_h > r_{0,h}} \|\Pi_{\pm,h} W_\omega(\boldsymbol{\xi}, t)\|^2, \qquad N_0(t) = h_x h_y \sum_{R_h \le r_{0,h}} \|W_\omega(\boldsymbol{\xi}, t)\|^2. \tag{S110}$$

The neutral sector N_0 contains numerically zero-symbol modes, not a third spin orientation. On the same stored field, $N(t) = N_+(t) + N_-(t) + N_0(t) = N_D(t)$. The absolute component rates and their sum are

$$g^{\mathrm{br}}_{\alpha,h}(t) = \frac{\kappa N_\alpha(t)}{P_0}, \qquad \sum_{\alpha\in\{+,-,0\}} g^{\mathrm{br}}_{\alpha,h}(t) = J_{\rho,h}(t). \tag{S111}$$

For positive denominators, the instantaneous and window-integrated fractions are

$$w^{\mathrm{br}}_\alpha(t) = \frac{N_\alpha(t)}{N(t)}, \qquad \overline{w}^{\mathrm{br}}_\alpha(T) = \frac{\int_0^T N_\alpha(t)\,dt}{\int_0^T N(t)\,dt}, \qquad \alpha \in \{+,-,0\}. \tag{S112}$$

Each triplet sums to one; the integrated fractions weight times by boundary density and include the neutral sector. The same projectors resolve the spin factor in Eq. (S94): on nonneutral modes with nonzero density,

$$a_\omega(\boldsymbol{\xi}, t) = -2\frac{\|\Pi_{+,h} W_\omega\|^2 - \|\Pi_{-,h} W_\omega\|^2}{\|W_\omega\|^2}. \tag{S113}$$

The covariance retains the momentum- and time-resolved information in this expression, which is lost in the integrated branch fractions.

Particle-spin polarization. For normalized initial data, we define the detector-associated spin-matrix rate $G_\omega(t)$ and the subnormalized reduced spin matrix $Q_\omega(t)$ of the state surviving no click up to time t, in continuum notation

$$G_\omega(t) = \kappa \int_{\Sigma_L} \Psi_\omega \Psi_\omega^\dagger \, dA, \qquad Q_\omega(t) = \int_{\Omega_\omega} \Psi_\omega \Psi_\omega^\dagger \, dV. \tag{S114}$$

Their traces are $g(t;\omega)$ and $S(t;\omega)$, respectively. The diagonal entries of G_ω resolve the detector rate in the chosen laboratory spin basis. The corresponding Pauli polarizations are

$$P_i^{\mathrm{det}}(t) = \frac{\mathrm{tr}[\sigma_i G_\omega(t)]}{g(t;\omega)}, \qquad P_i^{\mathrm{nc}}(t) = \frac{\mathrm{tr}[\sigma_i Q_\omega(t)]}{S(t;\omega)}, \qquad i \in \{x, y, z\}. \tag{S115}$$

Here "nc" denotes conditioning on no click; $P_z = +1$ is the prepared z-spin orientation. These quantities characterize particle-spin content within the model without assuming an additional apparatus spin measurement. The detector polarization accumulated over the window is

$$\overline{P}_i^{\mathrm{det}}(T) = \frac{\int_0^T \mathrm{tr}[\sigma_i G_\omega(t)]\,dt}{D_T(\omega)}. \tag{S116}$$

Numerical evaluation. Detector polarizations use accepted-step CN event matrices, with interval overlap for window integrals and step-midpoint plotting times. Survivor polarizations use the recorded diagnostic-time matrices. Each ratio uses the trace of the same matrix record; integrated polarizations use the trace of the integrated matrix. Branch integrals use piecewise-linear histories on their actual recording times. Undefined ratios are omitted; no smoothing or additional signal threshold is applied.

TABLE S4. Integrated branch fractions and particle-spin polarizations at $T = 40$. Bars denote ratios of integrals over $[0, 40]$. Branch fractions include the neutral sector and use native boundary-density records; detector polarizations use accepted-step event matrices. $P_z^{\mathrm{nc}}(40)$ describes the entire final surviving state. Displayed digits do not specify discretization accuracy; rounded fractions need not sum exactly to one.

ω	$\overline{w}_+^{\mathrm{br}}(40)$	$\overline{w}_-^{\mathrm{br}}(40)$	$\overline{w}_0^{\mathrm{br}}(40)$	$\overline{P}_z^{\mathrm{det}}(40)$	$P_z^{\mathrm{nc}}(40)$
1	0.442	0.430	0.128	0.907	0.940
25	0.819	0.114	0.067	0.307	0.998
50	0.879	0.073	0.049	0.232	0.999
75	0.902	0.058	0.040	0.193	1.000
100	0.916	0.049	0.035	0.169	1.000
150	0.931	0.040	0.029	0.143	1.000
200	0.938	0.036	0.026	0.128	1.000
300	0.946	0.032	0.022	0.097	1.000

Results and interpretation. At $\omega = 1$, the integrated nonneutral branch fractions are nearly balanced despite substantial instantaneous variation. At $\omega = 300$, the + branch accounts for approximately 95% of the integrated boundary density in both diagnostic windows, $[0, 3.6]$ and $(3.6, 40]$. The detected and surviving ensembles have different laboratory-z spin content: at $\omega = 300$, $\overline{P}_z^{\mathrm{det}}(40) \simeq 0.097$, while $P_z^{\mathrm{nc}}(40)$ is close to unity. The former averages detections over the window; the latter describes the entire state surviving at its endpoint. The integrated detector z-polarization decreases across the sampled confinements. Branch fractions resolve momentum-dependent transverse spin axes, whereas P_z uses a fixed laboratory axis. Unequal weights in the evolved boundary trace are compatible with equal local absorption probabilities for equally normalized incident branches at fixed normal and tangential wave numbers in the matching problem of main-text Sec. 4(ii).

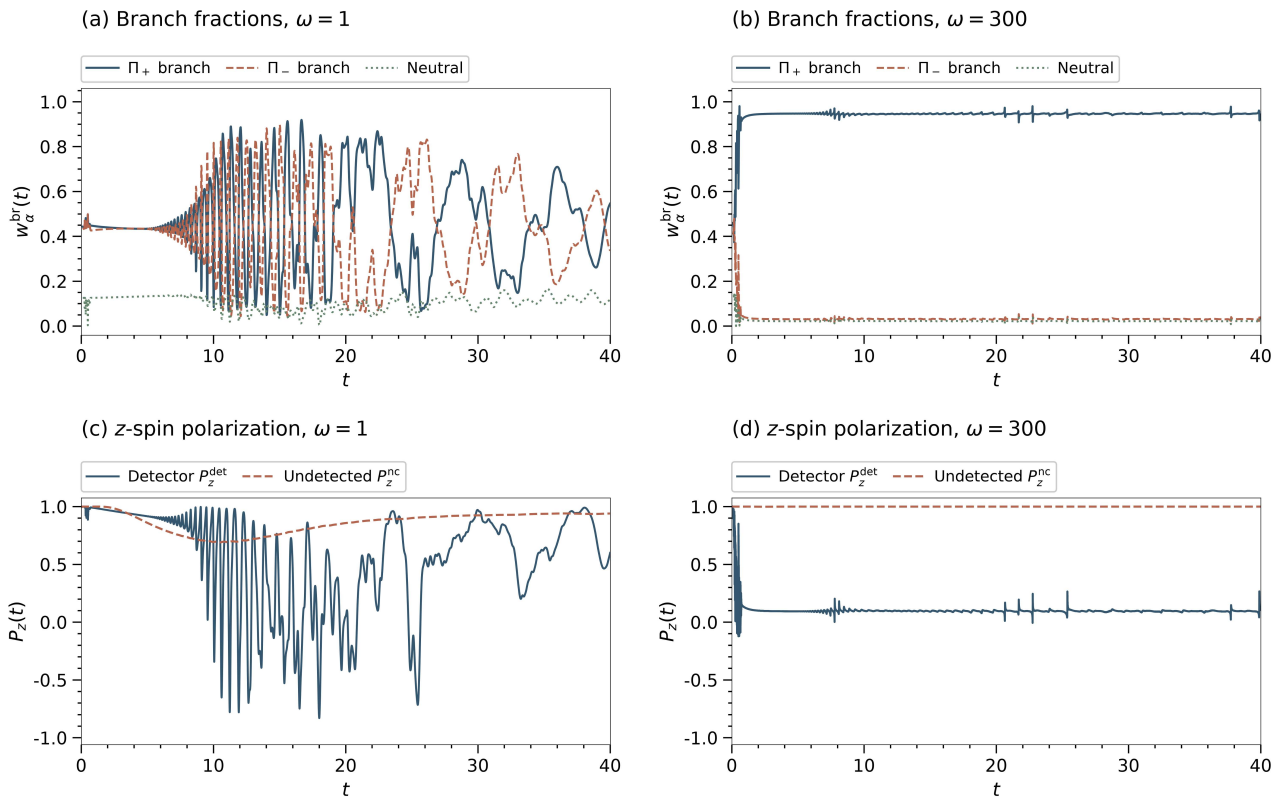


FIG. S1. Time-dependent branch fractions and particle-spin polarizations for the reference preparation of Table S1, $\eta = (1,0)^{\mathsf{T}}$, $k_z = \kappa = \pi$, $L = 20$, and $\sigma_{\parallel} = 0.5$. Left: $\omega = 1$; right: $\omega = 300$. (a),(b) Boundary fractions in the $+$, $-$ and neutral sectors. (c),(d) Detector-associated and whole-survivor laboratory z-spin polarizations. Curves connect defined recorded ratios without smoothing; the absolute detector rates are shown in main-text Fig. 1.

S6.3. Refinement checks at $\omega = 300$ and $T = 40$

This four-run series tests separate transverse, longitudinal, and time-step refinements with the reference preparation (Table S5). The baseline and spatial refinements use $\Delta t = 2 \times 10^{-4}$. The numerical detector and lateral-wall positions are held fixed, and the longitudinal refinement halves h_z. All statistics below use $T = 40$. For baseline b, refinement r, and a comparison window $W \subseteq [0, T]$, define the

following percentage discrepancies:

$$\epsilon_{\mathrm{rel}} = \frac{\left|\mu^{*,(r)}(T) - \mu^{*,(b)}(T)\right|}{\left|\mu^{*,(r)}(T)\right|} \times 100\%,$$
$$\epsilon_{\mathrm{CN}}(W) = \frac{\int_W \left|g_{\mathrm{CN}}^{(r)}(t) - g_{\mathrm{CN}}^{(b)}(t)\right| dt}{\int_W g_{\mathrm{CN}}^{(b)}(t)\, dt} \times 100\%, \tag{S117}$$
$$\epsilon_{\mathrm{br}}(W) = \frac{\sum_{\alpha\in\{+,-,0\}} \int_W \left|g_{\alpha,h}^{\mathrm{br},(r)}(t) - g_{\alpha,h}^{\mathrm{br},(b)}(t)\right| dt}{\int_W J_{\rho,h}^{(b)}(t)\, dt} \times 100\%.$$

Here ϵ_{rel} uses the *refined* mean in the denominator. The accepted-step rate g_{CN} is the recorded detector probability divided by the step duration and launch norm. The sampled branch rates $g_{\alpha,h}^{\mathrm{br}}$, including the neutral sector, sum to $J_{\rho,h}$ as defined above; all branch differences use this same total baseline denominator. CN rates are piecewise constant on accepted steps, while sampled branch densities are interpolated linearly on their native times. Absolute differences are integrated on the union of the time grids, splitting zero crossings of the linear differences; no smoothing or time shift is applied.

TABLE S5. Spatial and time-step refinement checks at $\omega = 300$ and $T = 40$, with the reference preparation of Table S1 and fixed numerical boundary positions. (a) Detection statistics, integrated $+-$branch fraction, and full-window rate discrepancies. (b) Integrated detector and final whole-survivor z-polarizations, boundary-weighted momentum moment and covariance, and late-window rate discrepancies. Discrepancies are defined in Eq. (S117); unmarked rate discrepancies use $[0, 40]$, and late discrepancies use $[8, 40]$. All ϵ entries are percentages. Displayed digits do not specify discretization accuracy.

(a) Refinement	$N_x = N_y$	N_z	Δt	D_T	$\mu^*(T)$	ϵ_{rel}	$\bar{w}_+^{\mathrm{br}}(T)$	ϵ_{CN}	ϵ_{br}
Baseline	100	1500	2×10^{-4}	0.739	18.631	0	0.946	0	0
Transverse	120	1500	2×10^{-4}	0.742	18.510	0.654	0.946	0.742	0.752
Longitudinal	100	2999	2×10^{-4}	0.733	18.977	1.823	0.952	6.188	6.320
Time step	100	1500	1×10^{-4}	0.740	18.626	0.031	0.946	2.647	2.647

(b) Refinement	$N_x = N_y$	N_z	Δt	$\overline{P}_z^{\mathrm{det}}(T)$	$P_z^{\mathrm{nc}}(T)$	$\mathbb{E}_{\nu_\omega^{(0)}}[s^2]$	$10^3\beta_\omega^{\mathrm{bl}}$	$\epsilon_{\mathrm{CN}}^{\mathrm{late}}$	$\epsilon_{\mathrm{br}}^{\mathrm{late}}$
Baseline	100	1500	2×10^{-4}	0.097	1.0	5.524	1.47	0	0
Transverse	120	1500	2×10^{-4}	0.100	1.0	5.607	0.99	0.588	0.610
Longitudinal	100	2999	2×10^{-4}	0.104	1.0	6.167	14.856	9.580	9.857
Time step	100	1500	1×10^{-4}	0.097	1.0	5.524	1.661	5.332	5.314

All three refinements in Table S5 preserve the integrated $+$-branch predominance and the contrast between detector-associated and final in-guide z-polarizations. Small changes in the restricted mean

do not imply comparably small changes in the detector and branch-rate histories, particularly in the late window. Longitudinal refinement produces the largest changes in these histories and the boundary-weighted momentum moment, with a particularly strong change in the covariance. The principal branch and spin contrasts therefore survive the tested refinements, while detailed profiles and auxiliary diagnostics remain resolution-sensitive.

S7. REDUCED RETURN CLOSURE

Section S6 characterizes the signal at the detecting plane. Here we separate exact probability balance near that plane from an assumed scalar description of later returns of undetected amplitude. The physical detection density remains $g(t;\omega) = J_L(t)$.

Exact probability balance. For $0 < \ell < L$, introduce a matching plane and the adjoining control region within the original guide:

$$\Sigma_* = \{z = z_*\}, \qquad z_* = L - \ell, \qquad \Omega_{\mathrm{bl}}^{(\ell)} = \{\mathbf{r} \in \Omega_\omega : z_* < z < L\}. \tag{S118}$$

Its probability content is

$$P_{\mathrm{bl}}^{(\ell)}(t) = \int_{\Omega_{\mathrm{bl}}^{(\ell)}} \rho(\mathbf{r}, t)\, dV. \tag{S119}$$

Define the matching-plane and detector fluxes by

$$J_*(t) = \int_{\Sigma_*} j_z(x, y, z_*, t)\, dx\, dy, \qquad J_L(t) = \int_{\Sigma_L} j_z(x, y, L, t)\, dx\, dy, \tag{S120}$$

where j_z is the normal component of the Pauli current and $J_* > 0$ means net flow into the control region from below. The spinor ABC gives

$$J_L(t) = \kappa \int_{\Sigma_L} \rho(x, y, L, t)\, dx\, dy \geq 0. \tag{S121}$$

With no lateral flux, continuity gives the exact balance

$$\frac{d}{dt} P_{\mathrm{bl}}^{(\ell)}(t) = J_*(t) - J_L(t). \tag{S122}$$

Probability entering the region can be detected, remain temporarily stored, or leave again through Σ_*. The signed net flux J_* does not separate these transfers into individual encounters.

Assumed return closure. Let $F_{\mathrm{out}}^{\mathrm{nc}}(t;\omega,\ell)$ denote the unresolved outgoing flux of undetected amplitude through Σ_*. A causal kernel $k_{\omega,\ell}^{\mathrm{ret}}(u)$ represents its later contribution to the detector signal, where $u \geq 0$ is the lag and $t - u$ the earlier departure time. The branch fractions of Sec. S6.2 describe spin–momentum composition rather than separate encounters; neither they nor the early/late split of Sec. S3 determines $F_{\mathrm{out}}^{\mathrm{nc}}$ or this kernel. We write

$$g_{\mathrm{box}}(t;\omega) = g_{\mathrm{first}}(t;\omega) + \int_0^t k_{\omega,\ell}^{\mathrm{ret}}(u) F_{\mathrm{out}}^{\mathrm{nc}}(t - u;\omega,\ell)\, du. \tag{S123}$$

Here g_{box} is the finite-guide detector signal within the reduced description, and g_{first} its first-encounter contribution, including the boundary response. Equation (S123) compresses unresolved phase, branch, and mode information into a scalar kernel; it is not an exact TDSE decomposition.

For a one-channel moment calculation, we make the additional assumption of a scalar convolution closure with a causal lag-only kernel $k^{\mathrm{eff}}_{\omega,\ell}$, distinct from $k^{\mathrm{ret}}_{\omega,\ell}$:

$$\begin{aligned} g_{\mathrm{box}} &= g_{\mathrm{first}} + K^{\mathrm{eff}}_{\omega,\ell} g_{\mathrm{box}}, \\ (K^{\mathrm{eff}}_{\omega,\ell} f)(t) &= \int_0^t k^{\mathrm{eff}}_{\omega,\ell}(u) f(t-u)\, du. \end{aligned} \tag{S124}$$

The kernel summarizes the unresolved no-click channel; the occurrence of g_{box} on the right does not mean that detected probability returns. If $g_{\mathrm{first}} \in L^1(0,\infty)$ and $\|k^{\mathrm{eff}}_{\omega,\ell}\|_{L^1(0,\infty)} < 1$, the Neumann series

$$g_{\mathrm{box}} = (I - K^{\mathrm{eff}}_{\omega,\ell})^{-1} g_{\mathrm{first}} = \sum_{n=0}^{\infty} (K^{\mathrm{eff}}_{\omega,\ell})^n g_{\mathrm{first}} \tag{S125}$$

converges in $L^1(0,\infty)$.

Time moments of the closure. Assume finite absolute zeroth and first moments of both fluxes and of the effective kernel, and define

$$M_0[f] = \int_0^\infty f(t)\, dt, \qquad M_1[f] = \int_0^\infty t f(t)\, dt. \tag{S126}$$

The kernel moments are

$$m^{(0)}_{\mathrm{ret}} = \int_0^\infty k^{\mathrm{eff}}_{\omega,\ell}(u)\, du, \qquad m^{(1)}_{\mathrm{ret}} = \int_0^\infty u k^{\mathrm{eff}}_{\omega,\ell}(u)\, du. \tag{S127}$$

For $m^{(0)}_{\mathrm{ret}} \neq 0$, write

$$\overline{U}_{\mathrm{ret}}(\omega;\ell) = \frac{m^{(1)}_{\mathrm{ret}}}{m^{(0)}_{\mathrm{ret}}}. \tag{S128}$$

For a nonnegative kernel, $m^{(0)}_{\mathrm{ret}}$ is an effective return weight and $\overline{U}_{\mathrm{ret}}$ its mean lag. A detection-density interpretation additionally requires nonnegative fluxes with total weights at most one. In general, these are the zeroth moment and the normalized first moment of the assumed kernel.

For positive zeroth flux moments, set

$$\overline{\tau}_{\mathrm{first}} = \frac{M_1[g_{\mathrm{first}}]}{M_0[g_{\mathrm{first}}]}, \qquad \overline{\tau}_{\mathrm{box}} = \frac{M_1[g_{\mathrm{box}}]}{M_0[g_{\mathrm{box}}]}. \tag{S129}$$

The convolution identities are

$$M_0[K^{\mathrm{eff}}_{\omega,\ell} f] = m^{(0)}_{\mathrm{ret}} M_0[f], \tag{S130}$$

$$M_1[K^{\mathrm{eff}}_{\omega,\ell} f] = m^{(0)}_{\mathrm{ret}} M_1[f] + m^{(1)}_{\mathrm{ret}} M_0[f]. \tag{S131}$$

Applying them to Eq. (S124) gives

$$\overline{\tau}_{\mathrm{box}} = \overline{\tau}_{\mathrm{first}} + \frac{m^{(1)}_{\mathrm{ret}}}{1 - m^{(0)}_{\mathrm{ret}}}. \tag{S132}$$

For $m^{(0)}_{\mathrm{ret}} \neq 0$, this is equivalently

$$\overline{\tau}_{\mathrm{box}} = \overline{\tau}_{\mathrm{first}} + \alpha_{\mathrm{ret}}(\omega;\ell)\overline{U}_{\mathrm{ret}}(\omega;\ell), \qquad \alpha_{\mathrm{ret}}(\omega;\ell) = \frac{m^{(0)}_{\mathrm{ret}}(\omega;\ell)}{1 - m^{(0)}_{\mathrm{ret}}(\omega;\ell)}. \tag{S133}$$

This separates the effective returned weight from its lag in the nonnegative-kernel interpretation. It is a moment identity of the assumed closure, not a fit to the simulations. The local scale $R = \sqrt{\omega}s$ and the computed $\beta^{\mathrm{bl}}_{\omega}$ do not determine the kernel moments.

Connection to the finite-window statistic. For $D_T > 0$, Eq. (S14) combines detected-event timing with the probability still undetected at T. The recorded histories do not determine the untruncated means entering the closure. The fixed-source lower-wall comparison in Sec. S9.4 provides separate numerical evidence that the late detector response depends on the reflecting boundary. The closure therefore provides a conditional interpretation of possible return contributions, whereas the reported finite-window statistics come from the full spinor-ABC evolution.

S8. CONTROL CHECKS

This section tests the sensitivity of the confinement contrast to selected bulk-spin couplings and transverse preparations, and compares the spinor ABC with finite relevant absorbing layers. The comparisons concern the stated configurations and observation windows; they do not establish a universal confinement law or uniqueness of the detector model.

S8.1. Bulk spin and transverse-state controls

We compare $\omega = 1$ and 100 using $H = H_{\mathrm{waveguide}}(\omega) + H_a$, retaining the longitudinal preparation, spatial grid, spinor ABC, and initial spinor $\eta = (1, 0)^T$ of Sec. S2. The Hermitian additions in Table S6 act throughout the guide from $t = 0$, with $w(z) = 1$, and the transverse state is initially $\chi^{(\omega)}_{00}$.

To test whether the confinement contrast requires transverse ground-state preparation, we instead set $H_a = 0$ and use the excited chiral profile

$$\chi^{(\omega)}_{+} = \frac{\chi^{(\omega)}_{10} + i\chi^{(\omega)}_{01}}{\sqrt{2}} = \sqrt{\omega}(X + iY)\chi^{(\omega)}_{00}, \tag{S134}$$

restricted and normalized on the same grid. Here $X = x - L_x/2$, $Y = y - L_y/2$, and $p_j = -i\partial_j$.

Each tested control gives a smaller D_{20} and larger $\bar{\tau}^{\mathrm{cond}}_{20}$ and $\mu^*(20)$ at $\omega = 100$ than at $\omega = 1$, with late detections at both confinements. Relative to the same-ω reference, the uniform and radial terms

change $\mu^*(20)$ by at most approximately 1.2% in magnitude. The Rashba and Rashba–Dresselhaus terms increase it by approximately 3.0–6.0% across the four comparisons. The chiral preparation increases it by approximately 5.5% at $\omega = 1$ and 24.7% at $\omega = 100$. Thus the confinement contrast persists for these couplings and this excited preparation, although its magnitude remains dependent on the dynamics and initial state. These two-confinement tests do not establish a scaling law or independence from arbitrary transverse preparations.

TABLE S6. Controls at $T = 20$, with two confinement rows per control. From normalized, piecewise-linear survival histories we evaluate $D_{20} = 1 - S(20)$, $\mu^*(20)$, and the absolute late probability $D_L = S(3.6) - S(20)$. The conditional mean follows from Eq. (S15). Entries are rounded independently.

Control and added Hamiltonian	ω	D_{20}	$\bar{\tau}_{20}^{\mathrm{cond}}$	$\mu^*(20)$	D_L
Reference: $H_a = 0$	1	0.925	3.947	5.151	0.348
	100	0.724	5.548	9.534	0.307
Uniform: $H_a = \pi\sigma_x + \frac{1}{2}\sigma_z$	1	0.928	3.938	5.091	0.350
	100	0.724	5.549	9.536	0.307
Radial: $H_a = 0.2\sqrt{\omega}(X\sigma_x + Y\sigma_y)$	1	0.923	3.935	5.165	0.346
	100	0.723	5.556	9.554	0.307
Rashba: $H_a = 0.5(\sigma_x p_y - \sigma_y p_x)$	1	0.920	4.111	5.375	0.356
	100	0.707	5.690	9.883	0.306
Rashba–Dresselhaus: $H_a = 0.4(\sigma_x p_y - \sigma_y p_x) + 0.3(\sigma_x p_x - \sigma_y p_y)$	1	0.917	4.147	5.460	0.355
	100	0.710	5.663	9.818	0.307
Chiral: $H_a = 0$, profile $\chi_+^{(\omega)}$	1	0.914	4.060	5.435	0.348
	100	0.598	6.434	11.89	0.287

S8.2. Hamiltonian-layer mimics

We compare the spinor ABC with finite absorbing layers, first using selected spin-dependent additions and then calibrating the complex entrance response. For the layer calculations, the ABC is removed and the guide extends to a nominal Dirichlet backing wall at $z = 22$.[4] The source and initial spin are retained: $z_c = 10$, $k_z = \kappa = \pi$, $\sigma_\parallel = 0.5$, and $\eta = (1, 0)^T$. The transverse grid is unchanged, and $h_z = 20/1500$ gives $N_z = 1650$ in the extended guide. For $H = H_{\mathrm{Herm}} - iW_{\mathrm{loss}}$, with $W_{\mathrm{loss}} \geq 0$, the detection density is

$$g_{\mathrm{layer}}(t) = 2\langle \Psi_t, W_{\mathrm{loss}}\Psi_t\rangle = -\partial_t \|\Psi_t\|^2. \tag{S135}$$

[4] Related scalar ABC limits involving complex potentials and Dirichlet-backed layers are recently discussed in Ref. [22] of the main-text.

All calculations use $T = 20$. The spinor-ABC reference values in Table S7 and S8 are taken from Table S6. Layer statistics are evaluated from normalized, piecewise-linear survival histories, with $D_L = S(3.6) - S(20)$. When the launch sample is absent, normalization uses the first recorded norm, with $S = 1$ before that sample.

Generic layers. The controls in Table S7 use the absorption profile

$$\chi(z) = \tfrac{1}{2}[1 + \tanh((z - 20)/0.165)], \tag{S136}$$

whose entrance tail extends below $z = 20$. Write $K = \sigma_x p_y - \sigma_y p_x$, $R = \sqrt{K^2}$, and $Q_s = (I + sK/R)/2$ for $R > 0$ and $s = \pm 1$. The projector Q_s selects $K = sR$; its tangential Fourier symbol is Π_{-s}. At the stated strengths, the spin-dependent additions change the scalar-layer statistics only weakly. Their restricted means lie between approximately 6.37 and 6.51, compared with 9.534 for the spinor ABC at $\omega = 100$.

TABLE S7. Generic absorbing-layer controls at $\omega = 100$ and $T = 20$. The spinor-ABC reference is taken from Table S6. The layer calculations use the listed Hamiltonian additions in the extended guide without the spinor ABC, with $\Delta t = 4 \times 10^{-4}$. Statistics use the survival-history convention specified in the text. Entries are rounded independently and do not specify discretization accuracy.

Layer addition / reference	D_{20}	$\bar{\tau}_{20}^{\mathrm{cond}}$	$\mu^*(20)$	D_L
Spinor ABC reference	0.724	5.548	9.534	0.307
$-i40\chi I$	0.857	4.240	6.487	0.304
$\sigma_x - i40\chi I$	0.857	4.240	6.487	0.304
$\chi\sigma_x - i40\chi I$	0.857	4.241	6.487	0.304
$-i\chi(40I + 20\sigma_y)$	0.862	4.198	6.374	0.306
$0.5\chi K - i40\chi I$	0.857	4.241	6.487	0.304
$-i\chi(40I + 0.5RQ_+)$	0.856	4.251	6.514	0.304

Impedance-calibrated layers. We next use a uniform layer with a separately chosen complex potential in each tangential branch. The added Hamiltonian term is

$$H_{\mathrm{layer}} = \mathbf{1}_{[20,22)}(z) V_{\mathrm{imp}}, \qquad V_{\mathrm{imp}} = \sum_{s=\pm 1} V_s(R) Q_s, \qquad -\operatorname{Im} V_s \geq 0. \tag{S137}$$

For fixed branch s and longitudinal energy E in the auxiliary stationary problem, the Dirichlet-backed field is proportional to $\sin[q_s(22 - z)]$. Its derivative-to-value ratio at the entrance is

$$\begin{aligned} \Lambda_s(E, R) &= -q_s \cot(q_s \ell), \quad q_s^2 = 2[E - V_s(R)], \\ Z_s(R) &= i\kappa - sR, \qquad \ell = 2. \end{aligned} \tag{S138}$$

Here Z_s is the target ABC impedance. Central-exact calibration imposes $\Lambda_s(E_{\mathrm{cal}}, R) = Z_s(R)$ at $E_{\mathrm{cal}} = k_z^2/2$. Weighted-broadband calibration starts from this solution and reduces the normalized

squared complex-impedance mismatch over the positive-k Gaussian band, using flux weights proportional to $k|\widetilde{\phi}_0(k)|^2$ and enforcing passivity. The objective also penalizes the mismatch at E_{cal}, with dimensionless weight w_c relative to the normalized band contribution. We use $w_c = 2$ at $\omega = 1$ and 100, and additionally test $w_c = 0.5$ at $\omega = 1$. The potentials are calibrated up to $R_{\mathrm{cal}} = 4\sqrt{\omega}$ and held at their cutoff values for larger R. Functions of K are evaluated in the spectral basis of the centered transverse Dirichlet finite-difference operators. This calibration concerns the auxiliary stationary problem; the full transverse harmonic evolution also couples the branches.

TABLE S8. Finite-window statistics at $T = 20$ for the spinor ABC, scalar complex absorbing potential layer (CAP, $-i40\chi(z)I$), and impedance-calibrated layers. The absorbing layers have nominal thickness $\ell = 2$. The weighted-broadband cases are labelled by the central-energy penalty weight w_{c}. At $\omega = 1$, the broadband calculations use $\Delta t = 8 \times 10^{-4}$ and the remaining cases use 10^{-3}; all $\omega = 100$ calculations use 4×10^{-4}. Entries are rounded independently and do not specify discretization accuracy.

Model	ω	D_{20}	$\bar{\tau}_{20}^{\mathrm{cond}}$	$\mu^*(20)$	D_L
Spinor ABC reference	1	0.925	3.947	5.151	0.348
Inverse: central-exact	1	0.805	4.817	7.773	0.379
Inverse: weighted-broadband ($w_{\mathrm{c}} = 2$)	1	0.911	4.064	5.480	0.382
Inverse: weighted-broadband ($w_{\mathrm{c}} = 0.5$)	1	0.905	4.003	5.522	0.363
Spinor ABC reference	100	0.724	5.548	9.534	0.307
Scalar CAP	100	0.857	4.240	6.487	0.304
Inverse: central-exact	100	0.530	6.645	12.927	0.254
Inverse: weighted-broadband ($w_{\mathrm{c}} = 2$)	100	0.636	5.904	11.031	0.272

Comparison and interpretation. At $\omega = 100$, the central-exact and weighted-broadband layers give $\mu^*(20) = 12.927$ and 11.031, respectively, compared with 9.534 for the spinor ABC in Table S6. Their relative restricted-mean discrepancies are therefore approximately 35.6% and 15.7%. Both broadband variants at $\omega = 1$ also improve the restricted-mean agreement relative to central-exact calibration. Broadband matching thus improves this integrated comparison, but does not establish equivalence of the full detection-time laws.

At $\omega = 100$, the broadband layer has a smaller detected fraction than the centered ABC reference (0.636 versus 0.724) and a larger conditional mean (5.904 versus 5.548). Both differences increase its restricted mean through Eq. (S14). Figure S2 shows the time profiles for these model classes using common bins of width 0.04. The full profiles distinguish temporal differences that are not specified by the restricted mean alone.

For the uniform inverse layer,

$$J_{\mathrm{entrance}} = g_{\mathrm{layer}} + \partial_t P_{\mathrm{layer}}, \tag{S139}$$

where P_{layer} is the probability stored in $20 \le z < 22$. The entrance current and detection rate therefore differ while the stored probability changes. The complex entrance impedance is a useful calibration target, but matching the auxiliary stationary response does not reproduce the full confined-wavepacket signal by itself.

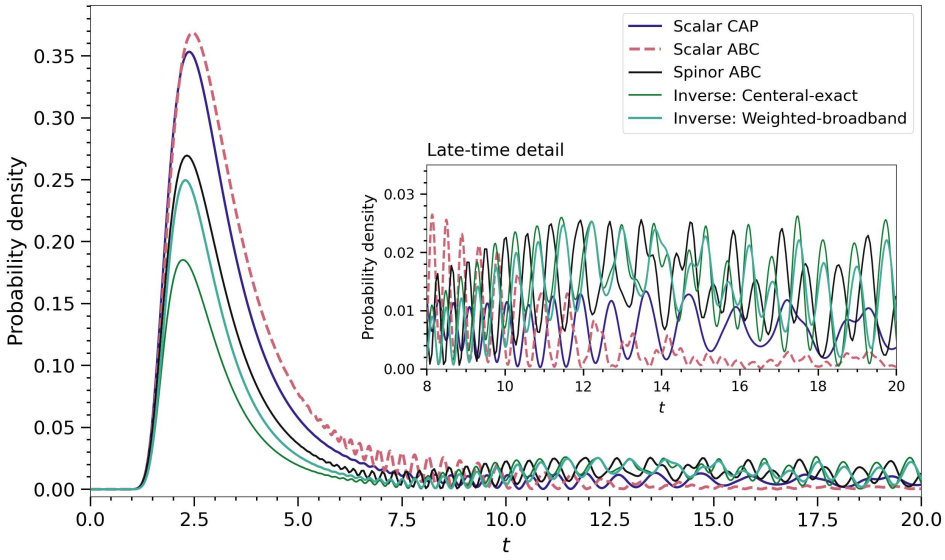


FIG. S2. Detection-time densities at $\omega = 100$ over $0 \le t \le 20$ for the spinor ABC, CAP, scalar ABC, central-exact inverse layer, and weighted-broadband inverse layer ($w_c = 2$), using common time bins of width 0.04. The inset resolves $8 \le t \le 20$. No time shift, amplitude rescaling, or conditioning on detection is applied.

S9. SOURCE, DETECTOR, AND GUIDE-LENGTH DEPENDENCE

We examine the dependence of the detector response on κ, k_z, $\sigma_\parallel$, and the guide geometry. Fixed-confinement scans characterize changes in the time profiles; separate confinement sweeps test the dependence beyond the reference preparation. We use the detection density and finite-window statistics defined in Secs. S2–S3. The parameter comparisons use $T = 20$, while the guide-length comparisons and fixed-source lower-wall controls use $T = 40$; these calculations are distinct from the main-text $T = 40$ reference sweep.

Where indicated, the confinement dependence is summarized by

$$\mu^*(T;\omega) \simeq A_T + B_T\sqrt{\omega}, \tag{S140}$$

with root-mean-square residual

$$r_{\mathrm{rms}} = \left[\frac{1}{n}\sum_{i=1}^{n}\left(\mu^*(T;\omega_i) - A_T - B_T\sqrt{\omega_i}\right)^2\right]^{1/2}. \tag{S141}$$

The coefficients depend on the preparation, geometry, horizon, and fitted confinement set. These fits summarize a finite range of the bounded statistic $\mu^*(T) \leq T$; r_{rms} measures fit agreement separately from numerical discretization accuracy.

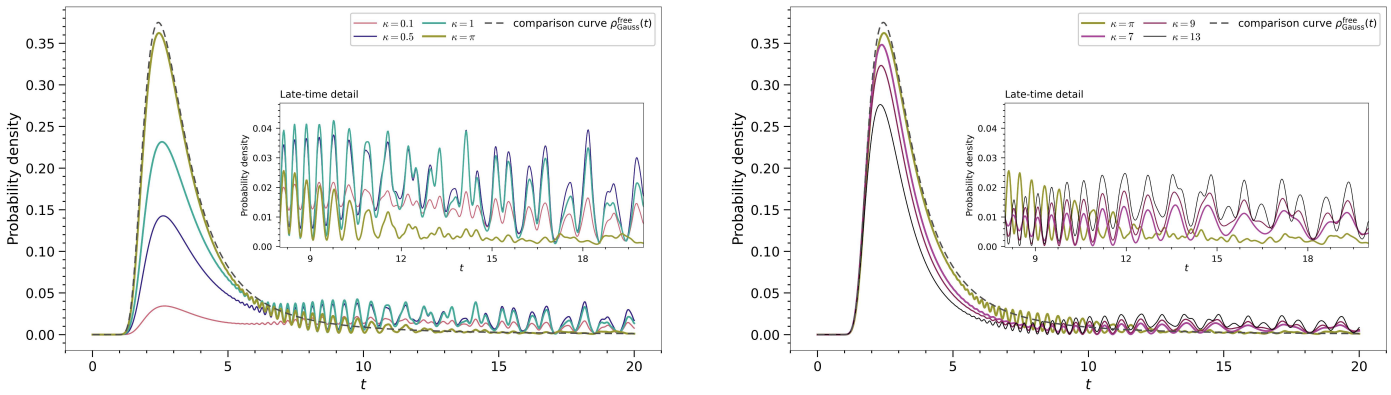
The fixed-confinement profiles at $\omega = 1$ in Figs. S3, S5, and S7 retain the earlier one-sided normal-boundary closure. At the checked parameter values, comparisons with the centered implementation showed no appreciable change in the qualitative trends. The confinement sweeps in Figs. S4, S6, and S8 use the centered implementation described in Sec. S2.

S9.1. Detector parameter κ

At $\omega = 1$, we vary κ with $k_z = \pi$ and $\sigma_{\parallel} = 0.5$. Figure S3 shows the time profiles, and Table S9 reports the restricted means at $T = 20$. Among the tabulated values, $\mu^*(20)$ is smallest at $\kappa = \pi$, the scalar matching point for the central longitudinal component [main-text Sec. 4(ii)].

TABLE S9. Finite-window statistic $\mu^*(T;\kappa)$ for the κ-scan at fixed $\omega = 1$, $k_z = \pi$, and $\sigma_{\parallel} = 0.5$ over $0 \leq t \leq 20$.

κ	0.1	0.5	1	2	π	5	7	9	13
$\mu^*(20;\kappa)$	17.016	11.709	8.247	5.653	5.151	5.729	6.717	7.687	9.343



(a) (b)

FIG. S3. Detector-time profiles over $0 \leq t \leq 20$ for the detector parameters κ indicated in the legends, at $\omega = 1$, $k_z = \pi$, and $\sigma_{\parallel} = 0.5$. The red dashed curves show the detector-free Gaussian comparison current for the fixed longitudinal preparation. The insets magnify the late-time signal.

The confinement sweeps at $\kappa = 2, \pi, 7$ all show increasing restricted means, while their ordering changes with confinement (Fig. S4). Thus the detector setting giving the smallest restricted mean at $\omega = 1$ need not retain that ordering throughout the sweep.

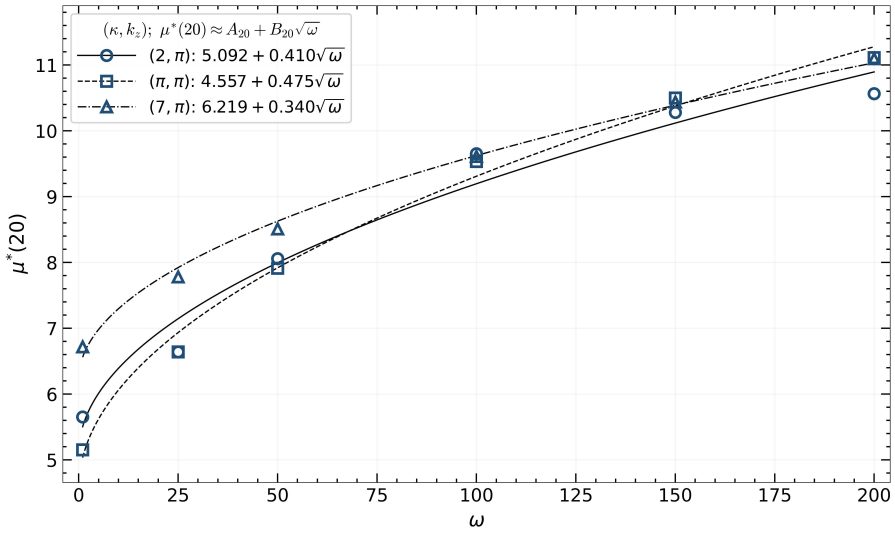


FIG. S4. Restricted means at $T = 20$ for $\kappa = 2, \pi, 7$, with $k_z = \pi$ and $\sigma_{\parallel} = 0.5$. Markers show the calculated values at $\omega = 1, 25, 50, 100, 150, 200$; lines show the empirical fits written in the legend, whose parameter pairs are ordered as (κ, k_z).

S9.2. Longitudinal wave number k_z

At $\omega = 1$, $\kappa = \pi$, and $\sigma_{\parallel} = 0.5$, Fig. S5 compares detector-time profiles for $k_z = 0.5, 1, 2, \pi, 5$ over $0 \le t \le 20$.

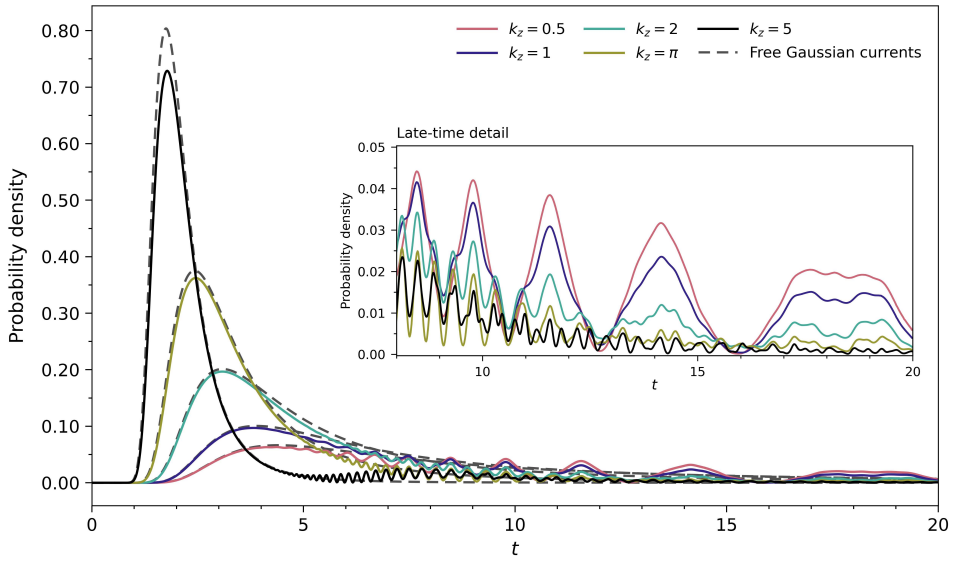


FIG. S5. Detector-time profiles over $0 \le t \le 20$ for $k_z = 0.5, 1, 2, \pi, 5$ at $\omega = 1$, $\kappa = \pi$, and $\sigma_{\parallel} = 0.5$. The red dashed curves are the detector-free Gaussian comparison currents for the corresponding preparations. The inset magnifies the late-time signal.

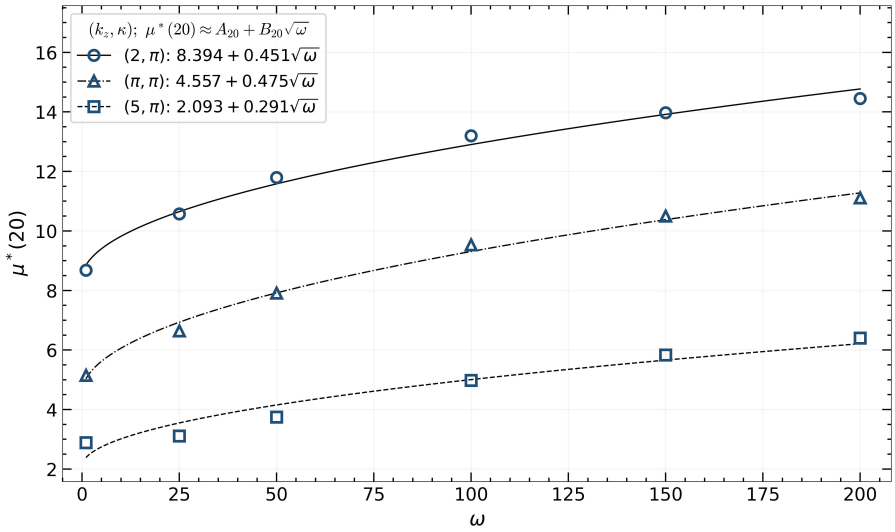


FIG. S6. Restricted means at $T = 20$ for $k_z = 2, \pi, 5$, with $\kappa = \pi$ and $\sigma_\parallel = 0.5$. Markers show the calculated values at $\omega = 1, 25, 50, 100, 150, 200$; lines show the empirical fits written in the legend, whose parameter pairs are ordered as (k_z, κ) in Table S10.

TABLE S10. Unweighted fits of $\mu^*(20; \omega)$ to $A_{20} + B_{20}\sqrt{\omega}$. The three rows correspond to the centered spinor ABC at fixed $\kappa = \pi$ in Fig. S6, using $\omega = 1, 25, 50, 100, 150, 200$. The residual r_{rms} is the root mean square of the six fit residuals.

(k_z, κ)	A_{20}	B_{20}	r_{rms}
$(2, \pi)$	8.394	0.451	0.212
(π, π)	4.557	0.475	0.179
$(5, \pi)$	2.093	0.291	0.335

S9.3. Longitudinal packet width $\sigma_\parallel$

At $\omega = 1$ and $k_z = \kappa = \pi$, Fig. S7 compares $\sigma_\parallel = 0.1, 0.25, 0.5, 0.75, 1$ over $0 \le t \le 20$. Changing this Gaussian amplitude width changes both spatial localization and the longitudinal momentum spread, $\Delta k_z = 1/(\sqrt{2}\,\sigma_\parallel)$. The narrower preparations show more pronounced oscillatory structure in these profiles. The restricted mean increases with confinement for all three widths tested in Fig. S8, with width-dependent empirical fit coefficients.

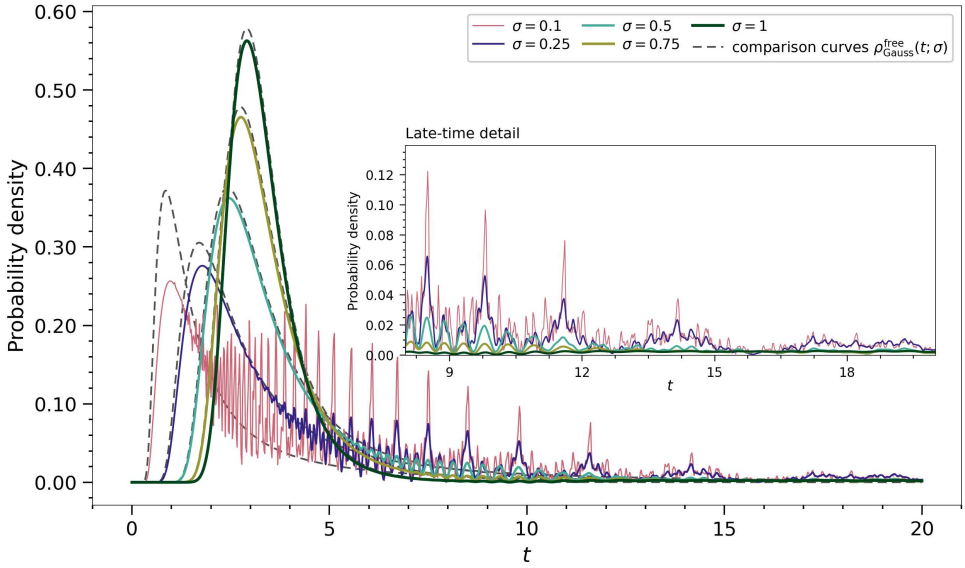


FIG. S7. Detector-time profiles over $0 \le t \le 20$ for $\sigma_{\parallel} = 0.1, 0.25, 0.5, 0.75, 1$ at $\omega = 1$ and $k_z = \kappa = \pi$. The widths refer to the Gaussian amplitude. Red dashed curves show the detector-free Gaussian comparison currents for the respective preparations; the inset magnifies the late-time signal.

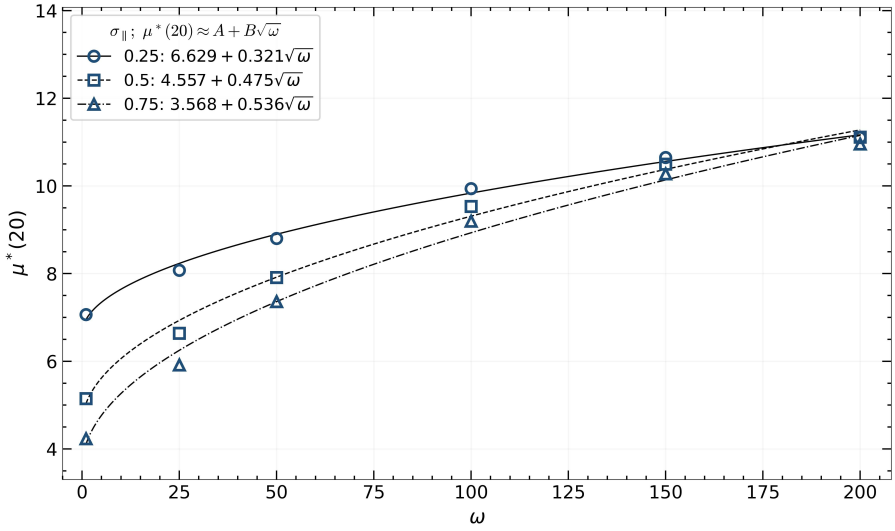


FIG. S8. Restricted mean detection times $\mu^*(20;\omega)$ for $\sigma_{\parallel} = 0.25, 0.5, 0.75$, with $L = 20$ and $k_z = \kappa = \pi$. Markers show the calculated values at $\omega = 1, 25, 50, 100, 150, 200$ for all three widths; lines show the empirical fits $A_{20} + B_{20}\sqrt{\omega}$ given in the legend.

S9.4. Guide geometry and lower-wall dependence

Midpoint-source comparison. Figure S9 shows representative detector-time profiles for the nominal $L = 30$ guide, with $z_c = L/2 = 15$, over $0 \le t \le 40$.

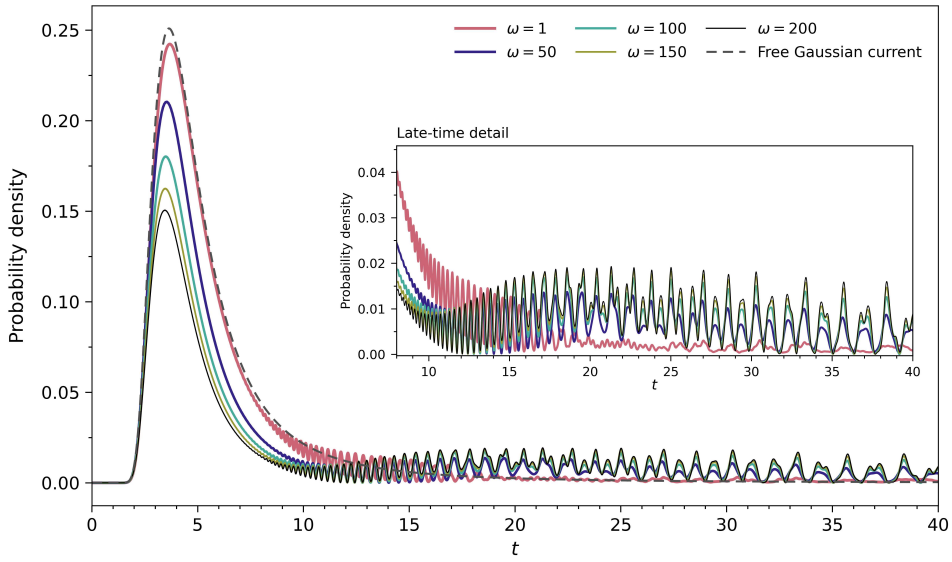


FIG. S9. Detector-time profiles for the nominal $L = 30$ guide, with $z_c = 15$, $k_z = \kappa = \pi$, and $\sigma_{\|} = 0.5$, over $0 \le t \le 40$. The displayed confinements are $\omega = 1, 50, 100, 150, 200$. Rates are reconstructed by numerical differentiation of the raw saved squared norms. The red dashed curve is the detector-free Gaussian comparison current; the inset magnifies the late-time signal.

Figure S10 compares the restricted means for nominal guide lengths $L = 20$ and 30 at the common horizon $T = 40$, with the source at $z_c = L/2$ in each guide.

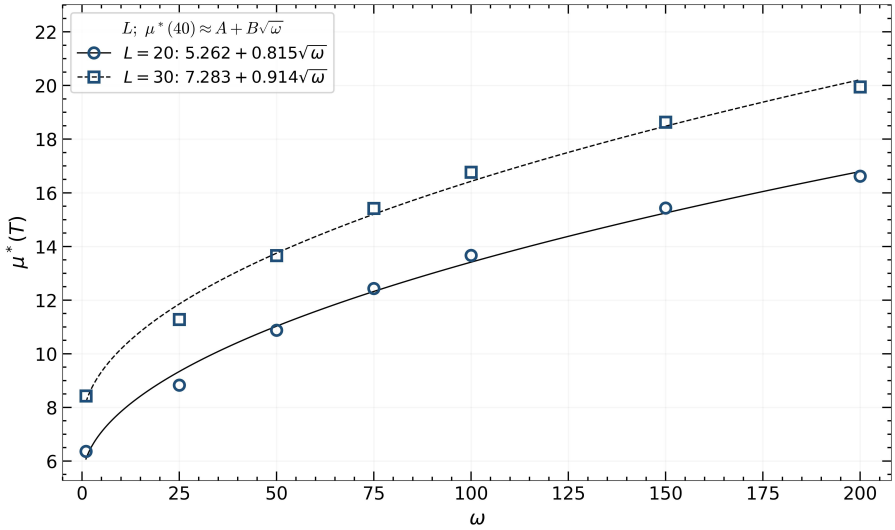


FIG. S10. Restricted means for nominal guide lengths $L = 20$ and 30 at the common horizon $T = 40$, with displayed markers at $\omega = 1, 25, 50, 75, 100, 150, 200$ for both series. Markers are calculated values and lines are unweighted fits to $A_{40} + B_{40}\sqrt{\omega}$. The source center is $z_c = L/2$.

Therefore the reported fits for $L = 20$ and 30 over $\omega \in \{1, 25, 50, 75, 100, 150, 200\}$ are

$$\mu^*(40;\omega) \simeq \begin{cases} 5.262 + 0.815\sqrt{\omega}, & L = 20, \quad r_{\mathrm{rms}} = 0.27, \\ 7.283 + 0.914\sqrt{\omega}, & L = 30, \quad r_{\mathrm{rms}} = 0.30. \end{cases} \tag{S142}$$

Fixed-source lower-wall control. We vary the reflecting lower-wall position at $\omega = 200$, keeping the source at $z_c = 10$ and the spinor-ABC detector at $z_D \simeq 19.986$, and evolve to $T = 40$. The preparation is $k_z = \kappa = \pi$, $\sigma_\parallel = 0.5$, and $\eta = (1, 0)^{\mathsf{T}}$. Moving the wall to $z_B = 0, -5, -10, -15$ increases N_z while preserving $h_z \simeq 0.013$, the transverse grid, and the original nonnegative-z nodes (Table S11).

TABLE S11. Fixed-source lower-wall controls at $\omega = 200$ and $T = 40$. Statistics are reported from normalized survival histories with the finite-window identity in Eq. (S14). Here $D_{L,40} = S(3.6) - S(40)$ and $t_{L,40}$ is the mean conditioned on detection in $(3.6, 40]$. Values are rounded independently.

z_B	N_z	D_{40}	$\bar{\tau}_{40}^{\mathrm{cond}}$	$\mu^*(40)$	$D_{L,40}$	$t_{L,40}$
0	1500	0.784	10.179	16.620	0.438	16.234
−5	1875	0.750	10.431	17.833	0.404	17.216
−10	2250	0.716	10.382	18.795	0.370	17.740
−15	2625	0.687	10.263	19.571	0.341	18.125

All four runs give $D_E \simeq 0.346$ at displayed precision. Moving the wall from $z_B = 0$ to -15 reduces D_{40} from 0.784 to 0.687 and $D_{L,40}$ from 0.438 to 0.341, while increasing the late conditional mean $t_{L,40}$ from 16.234 to 18.125. The overall detected conditional mean is nonmonotonic across the four positions. In the finite-window identity, the larger survival contribution $40S(40)$ more than compensates for the smaller detected-event contribution between these endpoints, raising the restricted mean from 16.620 to 19.571.

Figure S11 shows the corresponding signed longitudinal currents. These are finite-difference link currents, summed over the transverse grid and spin components and normalized by the initial norm, with positive direction toward the detector. The maps show net transport of undetected probability away from the upper region after the prompt signal. Their later evolution depends on the wall position.

The wall-displacement comparisons show lower-wall sensitivity of the late finite-window signal at the stated numerical settings, without identifying individual oscillations with particular round trips.

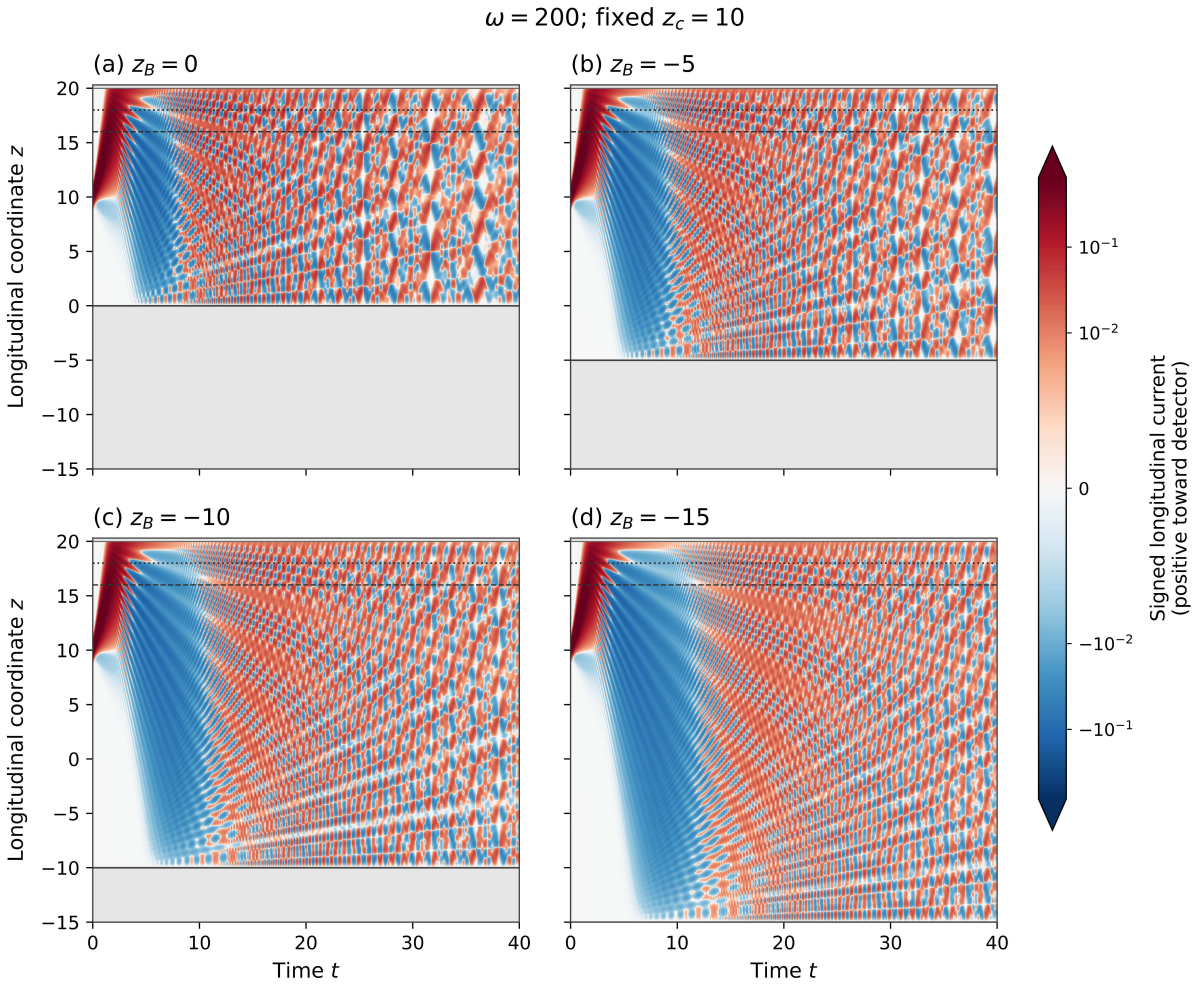


FIG. S11. Signed longitudinal-current maps at $\omega = 200$, over $0 \le t \le 40$, with fixed $z_c = 10$ and $z_D \simeq 19.9867$. Panels (a)–(d): $z_B = 0, -5, -10, -15$. Red/blue indicate net current toward/away from the detector; gray lies outside the guide. Dashed/dotted lines mark the monitors at $z \simeq 16.007, 18.007$.

S9.5. Absorbing lower-boundary control

We replace the lower Dirichlet wall by the componentwise scalar ABC

$$\partial_z \Psi(x, y, 0, t) = -i\kappa_{\text{lower}} \Psi(x, y, 0, t), \qquad \kappa_{\text{lower}} = \pi. \tag{S143}$$

The upper spin-coupled detector, bulk Hamiltonian, preparation, boundary positions, lateral walls, and spatial grid are retained. The controls use $\omega = 1, 200$, $T = 40$, and $\Delta t = 10^{-3}$, 2.5×10^{-4}, respectively. Both absorbing endpoints use centered ghost-point elimination and half longitudinal quadrature weight.

For incident normal wave number $k > 0$, the lower reflection probability is

$$\mathcal{R}_{\text{lower}}(k) = \left(\frac{k - \kappa_{\text{lower}}}{k + \kappa_{\text{lower}}} \right)^2 . \tag{S144}$$

For the ideal centered longitudinal stencil, k is replaced by $\sin(kh_z)/h_z$. The termination is therefore partially absorbing, not transparent to the complete wavepacket.

Upper-detection and lower-escape probabilities are accumulated separately from accepted-step events and satisfy

$$S_{\mathrm{in}}(t) + D_{\mathrm{top}}(t) + E_{\mathrm{lower}}(t) = 1, \tag{S145}$$

Here $S_{\mathrm{in}}(t)$ is the probability that neither upper detection nor lower escape has occurred by time t, whereas the probability of no upper click is $1 - D_{\mathrm{top}}(t) = S_{\mathrm{in}}(t) + E_{\mathrm{lower}}(t)$. The maximum recorded balance defect is below 4×10^{-7} in both runs; this checks accounting, not convergence. Total norm loss includes escape and is not the upper-click probability.

Figure S12 and Table S12 show that the partially absorbing lower boundary reduces late upper detections and their conditional mean time, while the early detected probability and mean time agree with the Dirichlet references at the reported precision. At $\omega = 200$, D_L decreases from 0.438 to 0.130 and t_L from 16.234 to 6.231. In this lower absorbing-boundary run, the window-integrated +-branch fraction and detector-associated z-polarization remain approximately 0.938 and 0.128, respectively, close to the Dirichlet values in Table S4; the final in-guide z-polarization remains close to unity. Thus the late signal changes substantially without removing these branch and spin contrasts. With the absorbing lower boundary, $D_{\mathrm{top}}(40)$ decreases from 0.905 at $\omega = 1$ to 0.476 at $\omega = 200$. Confinement-dependent suppression of upper detection therefore persists at these two frequencies without a perfectly reflecting lower wall. Residual oscillations remain, and the termination is only partially absorbing; these controls establish lower-boundary sensitivity but do not identify individual returns.

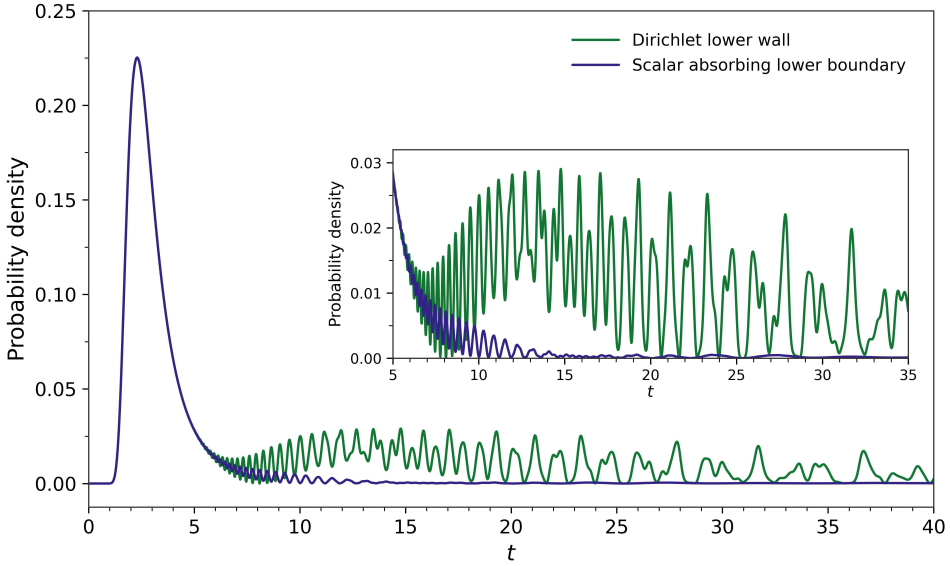


FIG. S12. Upper-detector click-time densities at $\omega = 200$ for a Dirichlet lower wall and a scalar absorbing lower boundary with $\kappa_{\text{lower}} = \pi$. Both retain the spin-coupled upper detector and the reference preparation of Table S1. Curves use accepted-step upper-boundary events without conditioning on detection; their areas equal $D_{\text{top}}(40)$. The inset resolves $5 \le t \le 35$.

TABLE S12. Lower-boundary controls at $T = 40$. Dirichlet rows reproduce Table S2, with $D_L = D_{40} w_L$ and $S_{\text{in}} = 1 - D_{40}$; scalar-ABC rows use accepted-step boundary events. Here $D_L = D_{\text{top}}(40) - D_{\text{top}}(3.6)$ and t_L are the probability and conditional mean time of upper detection in $(3.6, 40]$.

ω	Lower boundary	D_{top}	E_{lower}	S_{in}	$\bar{\tau}_{\text{top}}^{\text{cond}}$	D_L	t_L
1	Dirichlet	0.952	0	0.048	4.663	0.375	7.839
1	Scalar ABC	0.905	0.062	0.033	3.865	0.329	6.092
200	Dirichlet	0.784	0	0.216	10.178	0.438	16.234
200	Scalar ABC	0.476	0.472	0.052	3.525	0.130	6.231

S9.6. Long-time evolution at $\omega = 200$

To examine detection beyond the main observation window, we perform an additional centered spinor-ABC calculation at $\omega = 200$ up to $T = 200$. The physical preparation and spatial grid are those of Table S1, including the reflecting lower wall, with time step $\Delta t = 2.5 \times 10^{-4}$. Figure S13 shows an oscillatory detector signal extending well beyond $T = 40$, with late structure still visible in the interval $80 \le t \le 200$.

Estimates reconstructed from the logged survival history give $D_{40} \simeq 0.784$ and $D_{189.95} \simeq 0.941$. Thus, approximately 15.7% of the initial probability is detected in $(40, 189.95]$, corresponding to about 73%

of the probability still undetected at $T = 40$. The late tail therefore carries appreciable probability: finite-window survival should not be identified with permanent non-detection.

The time profile alone does not resolve individual encounters or establish progressive filtering over successive reflections. This single-confinement extension also leaves the unrestricted mean and confinement scaling at longer horizons undetermined.

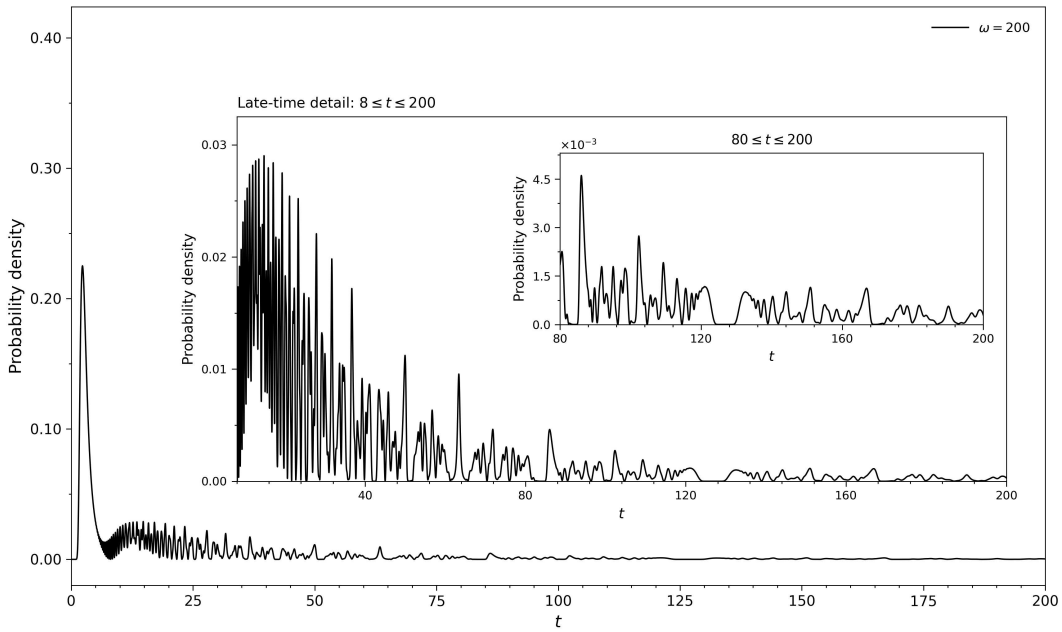


FIG. S13. Long-time detection-time profile over $0 < T < 200$ with $\omega = 200$. Using the centered spinor ABC and a reflecting lower boundary. The insets resolve the late time structure over $8 \le t \le 200$ and $80 \le t \le 200$.

S10. PHYSICAL SCALES

The analysis above is dimensionless. Let a_0 be the physical length represented by one dimensionless unit. Then

$$r_{\mathrm{phys}} = a_0 r, \qquad t_{\mathrm{phys}} = T_* t, \qquad T_* = \frac{m_{\mathrm{phys}} a_0^2}{\hbar}. \tag{S146}$$

The transverse trap frequency and longitudinal wave numbers scale as

$$\Omega_\perp = \frac{\omega}{T_*}, \qquad k_{z,\mathrm{phys}} = \frac{k_z}{a_0}, \qquad \kappa_{\mathrm{phys}} = \frac{\kappa}{a_0}. \tag{S147}$$

The physical transverse oscillator length is

$$\ell_{\perp,\mathrm{phys}} = \frac{a_0}{\sqrt{\omega}}. \tag{S148}$$

As an illustrative scale, for single ^{87}Rb atoms and $a_0 = 2\,\mu$m, one obtains

$$T_* \simeq 5.47\,\mathrm{ms}. \tag{S149}$$

Thus the observation windows $T = 20$ and $T = 40$ correspond to approximately 109 ms and 219 ms, respectively. The values $\omega = 1, 100, 300$ correspond to transverse oscillator lengths approximately 2.0 μm, 0.20 μm, and 0.115 μm, respectively.

A literal implementation of the spinor ABC would require an effective absorbing boundary whose entrance response contains the tangential spin–momentum branches

$$i\kappa \pm |\boldsymbol{\xi}|. \tag{S150}$$

Generic spin-dependent loss or bulk spin–orbit terms should not be expected to reproduce this response without engineering the corresponding boundary impedance. The layer comparisons in Sec. S8.2 use these branch impedances as calibration targets and quantify the resulting finite-window agreement. A microscopic physical realization remains outside the present study.